

Unit 13

AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY

118 items · 407 marks

Syllabus sections in this unit

- 13.1 Formulas, functional groups and the naming of organic compounds
- 13.2 Characteristic organic reactions
- 13.3 Shapes of organic molecules; σ and π bonds
- 13.4 Isomerism: structural isomerism and stereoisomerism

Unit 13: An introduction to AS Level organic chemistry**1. M/J 26/P21/Q5/a,c**

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

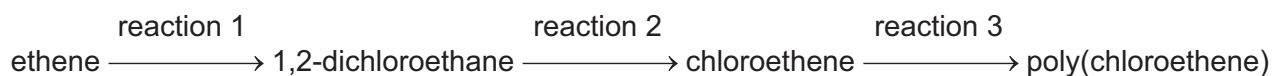


Figure 5.1

- (a)** Complete Table 5.1.

Table 5.1

compound	type of hybridisation of carbon atoms
1,2-dichloroethane	
chloroethene	

[2]

- (c)** Describe what is meant by heterolytic fission of a covalent bond.

[2]

2. M/J 26/P22/Q5/a,b(i,ii)

- 5 Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

(a) Complete Table 5.1. [2]

(b) The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

(i) Describe what is meant by free radical.

[1]

(ii) Name the type of bond breaking that occurs to produce the free radical $\cdot\text{Cl}$ from Cl_2 .

[1]

3. M/J 26/P22/Q6/a,b(ii)

6 (a) Figure 6.1 shows the structure of compound **P**.

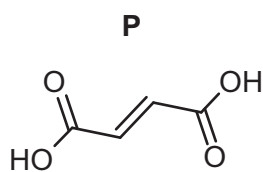


Figure 6.1

(i) Draw the structure of the stereoisomer of **P**.

[1]

(ii) Explain why **P** shows stereoisomerism.

[2]

(b) **P** reacts to produce compound **Q**.

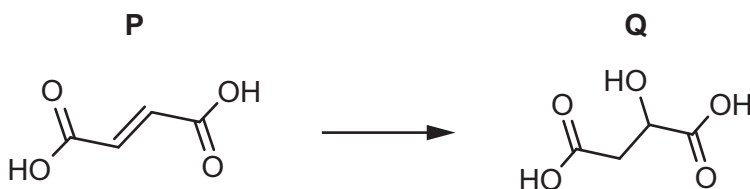


Figure 6.2

(ii) Complete Figure 6.3 to show the pair of stereoisomers of **Q**.

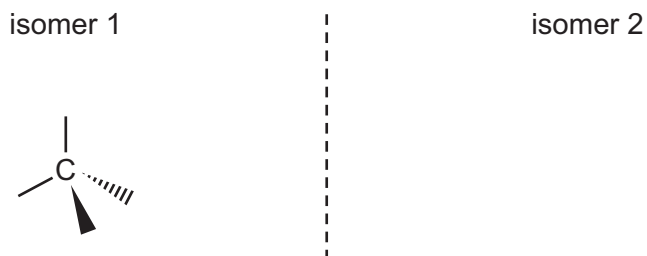


Figure 6.3

[2]

4. M/J 26/P23/Q4/c(i,ii)

4 The structure of compound **V** is shown in Figure 4.1.

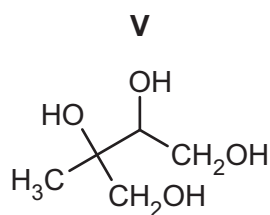


Figure 4.1

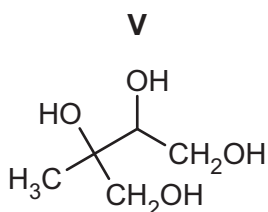


Figure 4.1

(c) Compounds **W**, **X** and **Y** are structural isomers of **V**.

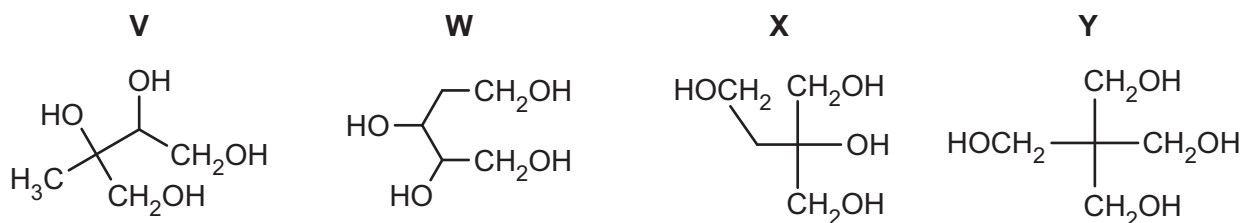


Figure 4.2

(i) Describe structural isomerism.

[2]

(ii) Identify which of structures **V**, **W**, **X** and **Y** show stereoisomerism.

Explain your answer by referring to the relevant parts of the molecule that cause stereoisomerism.

[2]

5. M/J 26/P23/Q5/a(i)

- 5 Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (a) (i) Complete Table 5.1.

[2]

6. M/J 26/P24/Q5/c

- 5 Long-chain alkanes from crude oil can be cracked to make alkenes.

- (c) But-1-ene and but-2-ene are structural isomers.

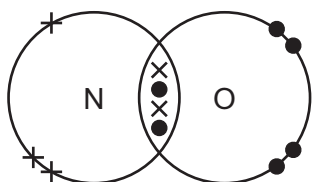
State the type of structural isomerism shown by but-1-ene and but-2-ene.

[1]

7. MAR 26/P22/Q2/a(i)

- 2 Nitrogen monoxide, NO , is an atmospheric pollutant.

- (a) Figure 2.1 shows the dot-and-cross diagram of NO .

**Figure 2.1**

- (i) NO has an unpaired electron.

State the term for a molecule with an unpaired electron.

[1]

8. MAR 26/P22/Q3/b

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

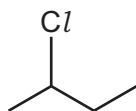
2-chlorobutane

Figure 3.1

(b) 2-chlorobutane can be formed by the electrophilic addition of HCl to different isomers of C_4H_8 . Compound **A** is one of these isomers.

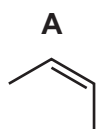
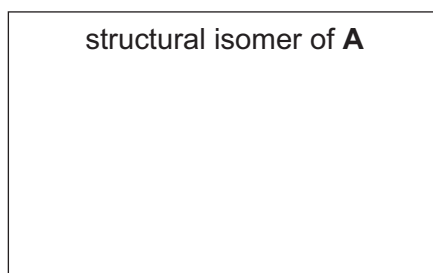


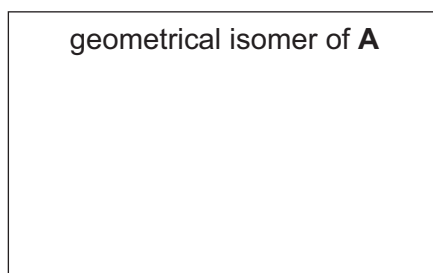
Figure 3.2

(i) Draw a structural isomer of **A** that can also form 2-chlorobutane when it reacts with HCl .



[1]

(ii) Draw and give the systematic name of the geometrical isomer of **A**.



systematic name

[2]

(iii) State and explain the origin of geometrical isomerism in alkenes.

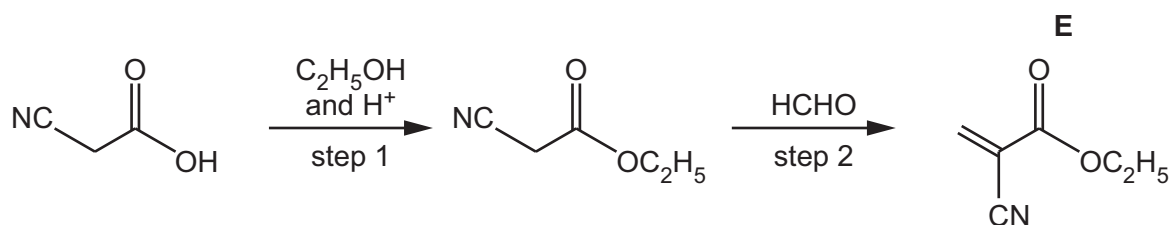
[2]

9. MAR 26/P22/Q4/d

also Unit 22 (2 marks), Unit 18 (1 mark), Unit 20 (1 mark); this unit 2 marks

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(d) Figure 4.4 describes a sequence of reactions, starting with $\text{CH}_2(\text{CN})\text{COOH}$, to form **E**.

**Figure 4.4**

(i) Identify the type of reaction that occurs in step 1.

[1]

(ii) Name HCHO , the reactant used in step 2.

[1]

(iii) Deduce the molecular formula of **E**.

[1]

(iv) **E** can be polymerised to form **F**.

Draw the structure of **one** repeat unit of polymer **F**.

[1]

(v) Figure 4.5 shows the infrared spectrum of **E**.

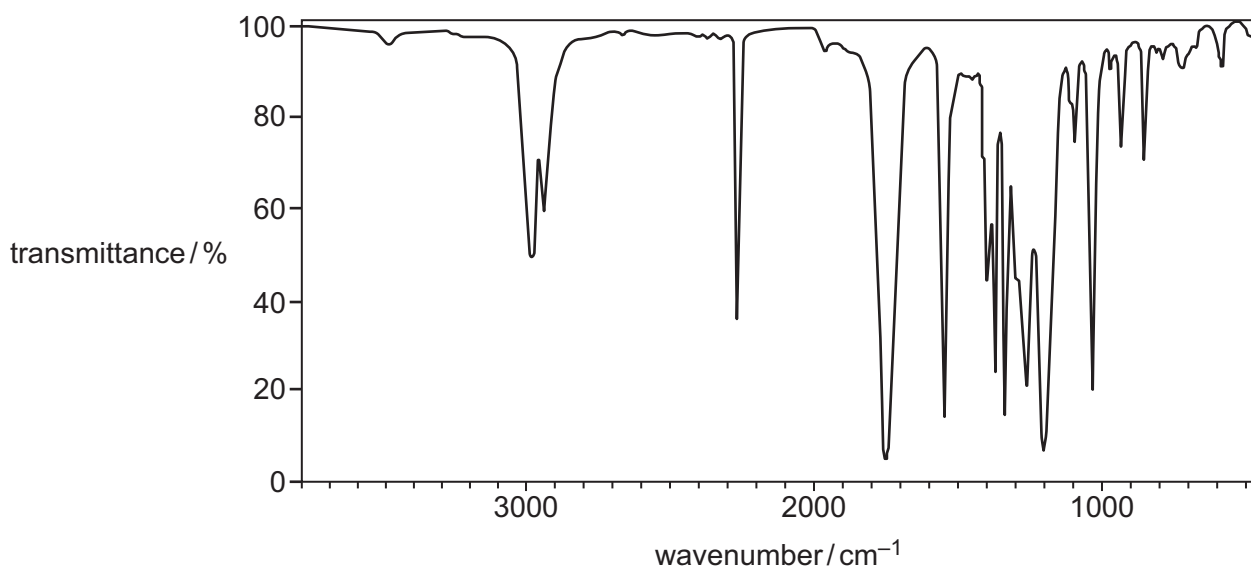
**Figure 4.5**

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Using the information given in Table 4.1, identify with a cross (X) on Figure 4.5 an absorption that can be used to monitor the progress of the polymerisation of **E** to form **F**.

Explain your answer with reference to the relevant bond.

[2]

10. O/N 25/P21/Q3/b,c(i)

- 3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C₅H₁₀, has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C₅H₁₀.

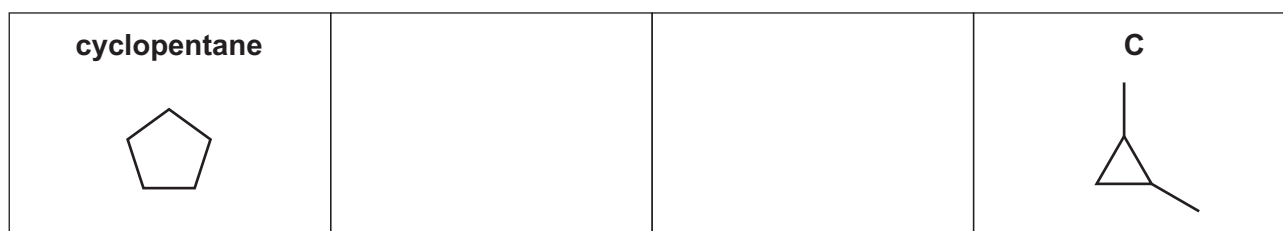


Fig. 3.1

[2]

(c) Cyclopentane reacts with Cl₂ in the presence of ultraviolet light to form C₅H₉Cl.

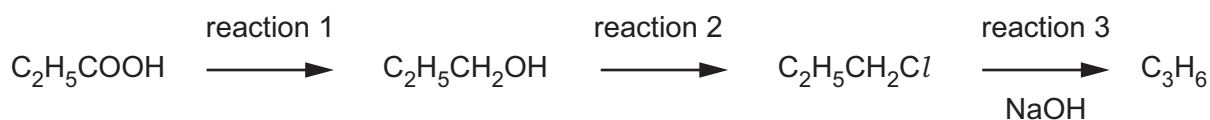
(i) The reaction is initiated by the bond fission of Cl₂.

State the type of bond fission shown in the initiation step.

[1]

11. O/N 25/P21/Q4/b(ii),c(i)

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(ii) Define stereoisomerism.

[2]

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



(i) Name **X**.

[1]

12. O/N 25/P22/Q3/a,d(i)

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

[2]

(d) Hexane, C_6H_{14} , has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

hexane	A	B	C	D

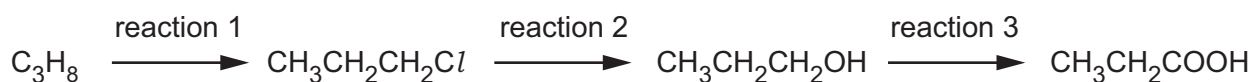
Fig. 3.1

(i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane.

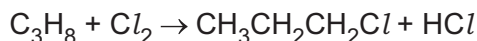
[2]

13. O/N 25/P22/Q4/a(iii,iv,v)

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



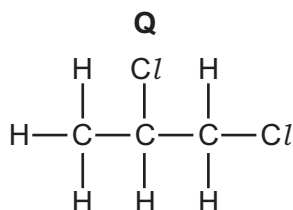
The reaction takes place via initiation, propagation and termination steps.

(iii) Reaction 1 is initiated by the bond fission of Cl_2 .

State the type of bond fission shown in the initiation step.

[1]

(iv) Compound **Q** is a by-product of reaction 1.



Name **Q**.

[1]

(v) The molecular formula of **Q** is $C_3H_6Cl_2$.

Identify the types of structural isomerism and stereoisomerism that a molecule with molecular formula $C_3H_6Cl_2$ can show.

type of structural isomerism

type of stereoisomerism

[2]

14. O/N 25/P23/Q3/b,c(i)

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C_5H_{10} , has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C_5H_{10} .

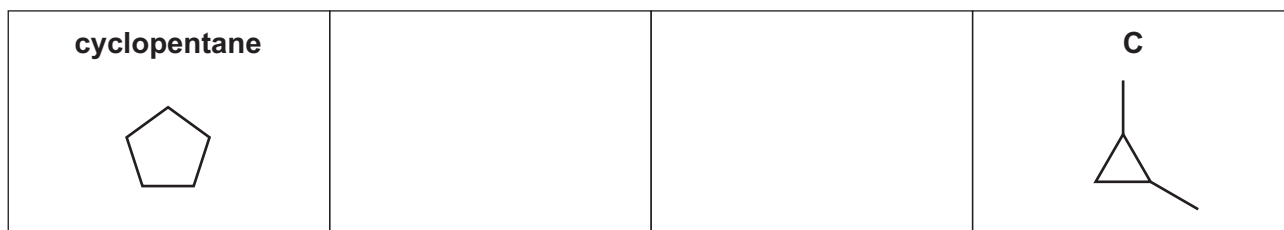


Fig. 3.1

[2]

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

(i) The reaction is initiated by the bond fission of Cl_2 .

State the type of bond fission shown in the initiation step.

[1]

15. O/N 25/P23/Q4/b(ii),c(i)

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

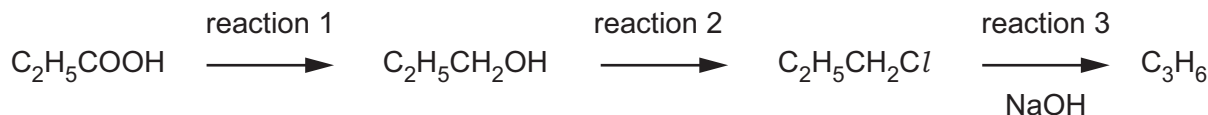


Fig. 4.1

(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(ii) Define stereoisomerism.

[2]

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



(i) Name **X**.

[1]

16. O/N 25/P24/Q3/a(iv)

- 3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

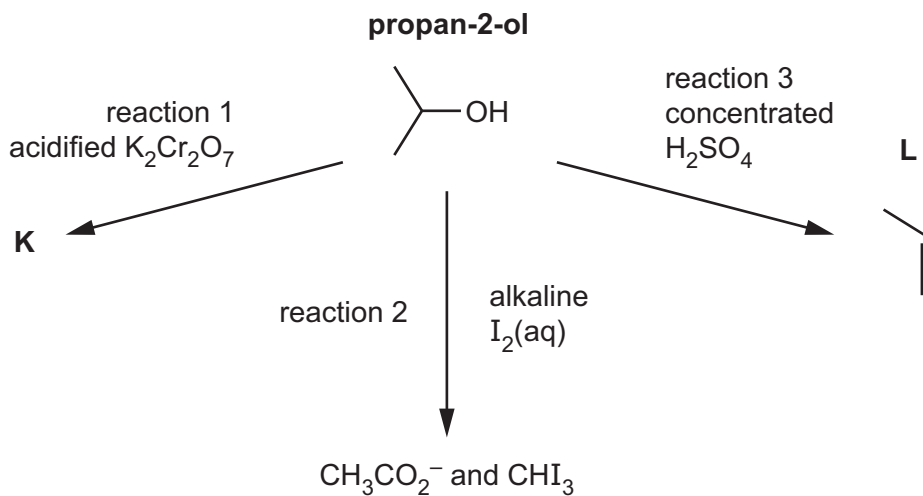


Fig. 3.1

- (iv) Complete Fig. 3.2 to show the pi (π) bond in **L** that is formed from orbital overlap.

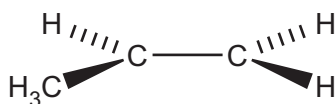


Fig. 3.2

[1]

17. M/J 25/P22/Q3/a(i)

- 3 (a) C_3H_6 reacts with $\text{HBr}(\text{g})$ in an addition reaction.

- (i) Define addition reaction.

[1]

18. M/J 25/P22/Q4/c(i)

- (c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

- (i) Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

[1]

19. M/J 25/P23/Q5/b(i)

5 (a) **W** is a colourless liquid.

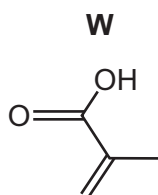


Fig. 5.1

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

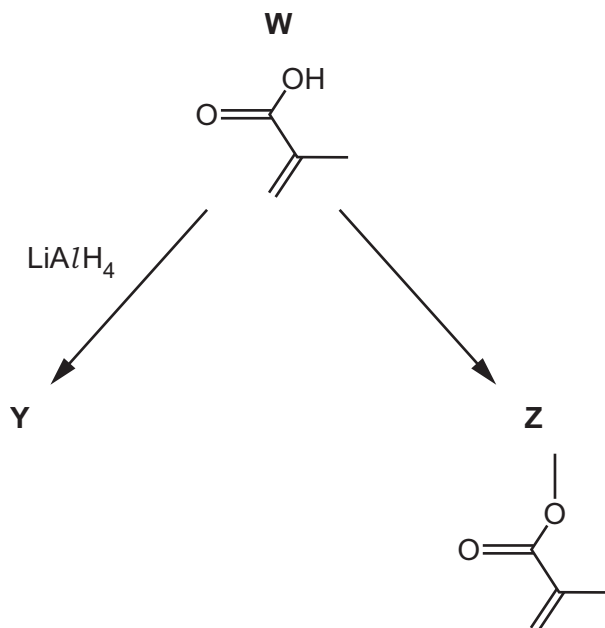


Fig. 5.2

- (i) Deduce the number of sigma (σ) bonds and pi (π) bonds present in **Z**.

number of σ bondsnumber of π bonds

[2]

20. M/J 25/P23/Q6/a(i)

6 (a) HOCl reacts with $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$ in an electrophilic addition reaction.

- (i) Define addition reaction.

[1]

21. M/J 25/P24/Q4/a

4 The skeletal formula of **E** is shown in Fig. 4.1.

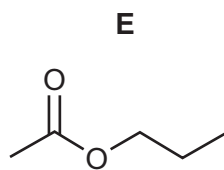


Fig. 4.1

(a) State the molecular formula of **E**.

[1]

22. M/J 25/P24/Q5/a,b,c

5 The skeletal formula of **W** is shown in Fig. 5.1.

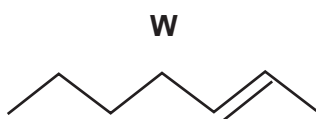


Fig. 5.1

(a) **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.



[2]

(b) (i) Describe the origin of stereoisomerism in **W**.

[1]

(ii) State why **W** shows stereoisomerism but one of the positional isomers in (a) does **not**.

[1]

(c) Draw the skeletal structure of the stereoisomer of **W**.

[1]

23. M/J 25/P24/Q6/a(iii)

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(iii) Molecules of **P** have straight chains.

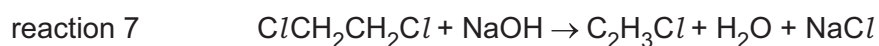
Name the homologous series to which **P** belongs.

[1]

24. MAR 25/P22/Q1/c(i)

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



(i) State what is meant by unsaturated.

[1]

25. MAR 25/P22/Q4/b(iii)

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

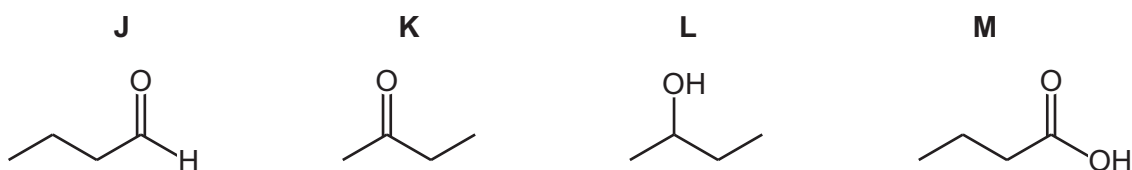


Fig. 4.1

(b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

(iii) **N** has a chiral centre.

Explain what is meant by a chiral centre.

[1]

26. O/N 24/P21/Q1/d(ii,iii,iv)

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

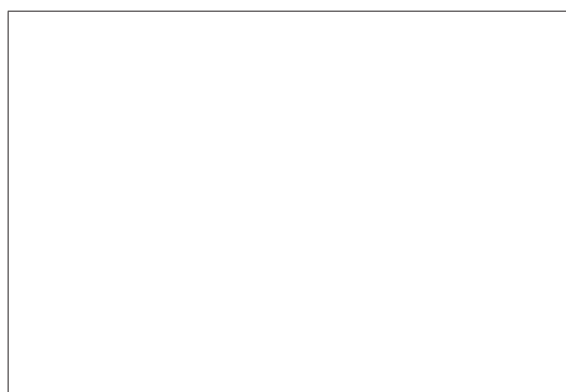
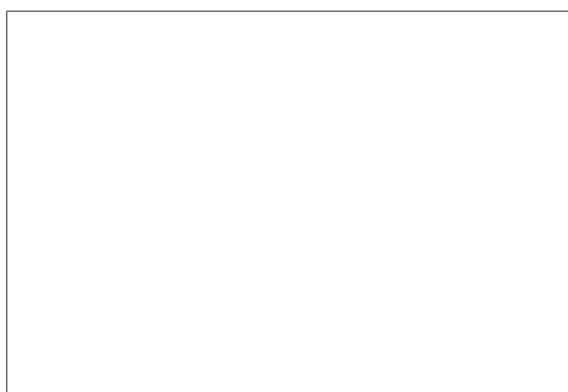
(ii) Define stereoisomers.

[1]

(iii) Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

(iv) Draw the two stereoisomers of but-2-ene.



[2]

27. O/N 24/P21/Q2/b(i)

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

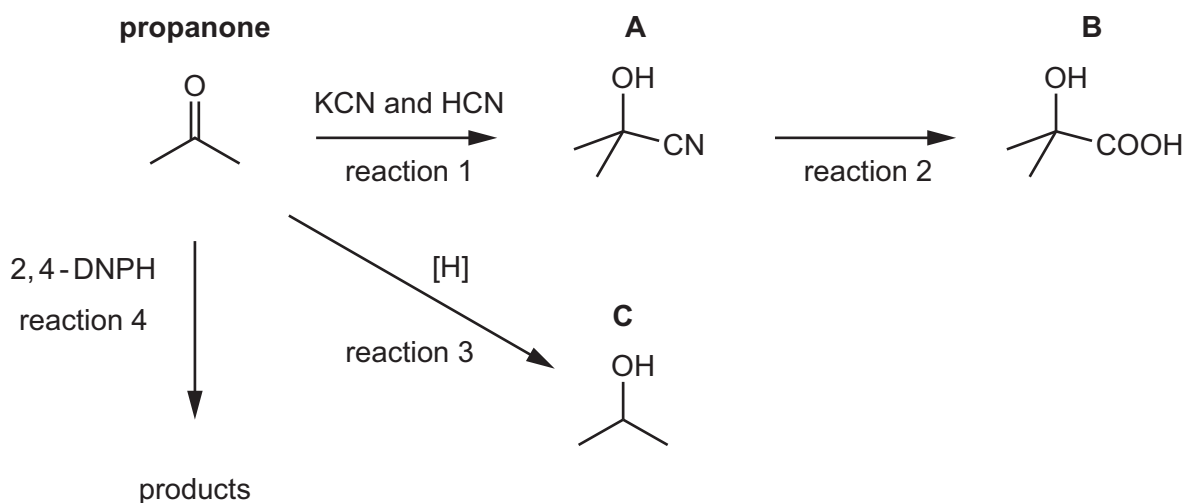
(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

(i) Define free radical.

[1]

28. O/N 24/P21/Q4/a(ii)

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (a) Reaction 1 is a nucleophilic addition reaction.
- (ii) Explain why **A** does **not** show optical isomerism.

[1]

29. O/N 24/P23/Q1/d(ii,iii,iv)

- 1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.
- (d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

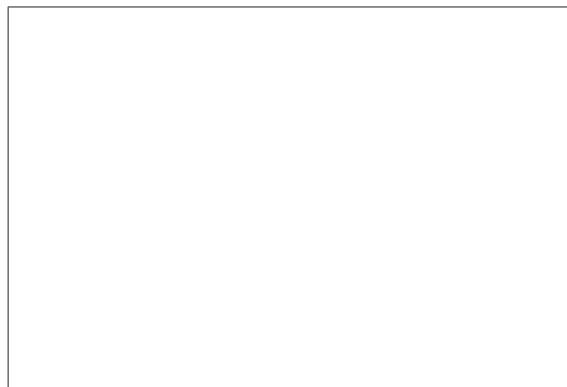
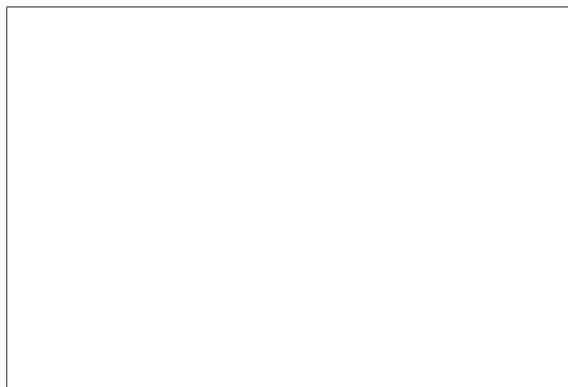
- (ii) Define stereoisomers.

[1]

- (iii) Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

- (iv) Draw the two stereoisomers of but-2-ene.



[2]

30. O/N 24/P23/Q2/b(i)

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

(i) Define free radical.

[1]

31. O/N 24/P23/Q4/a(ii)

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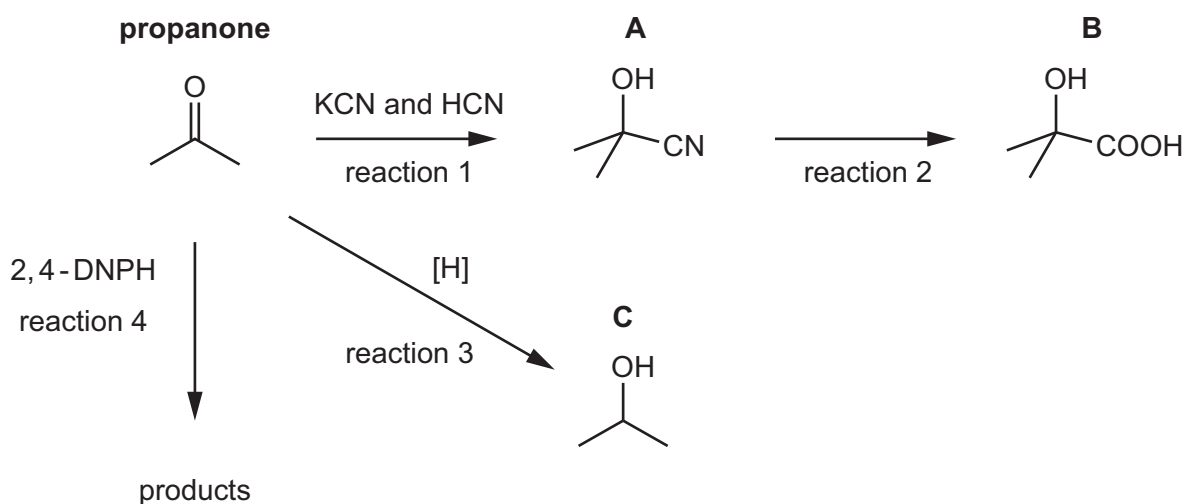


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(ii) Explain why **A** does **not** show optical isomerism.

[1]

32. M/J 24/P21/Q5/b,c(i)

5 Hydrocarbon molecules contain covalent bonds.

(b) A C=C bond in an alkene is made from a σ bond and a π bond.

(i) Identify the hybridisation of the carbon atoms in a C=C bond in an alkene.

[1]

(ii) Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a C=C bond.

σ bond

π bond

[2]

(c) In electrophilic reactions involving alkenes the π bond of C=C is broken.

(i) Suggest **one** difference between σ and π bonds that explains why the π bond is broken in electrophilic addition reactions involving alkenes.

[1]

33. M/J 24/P21/Q6/a

6 **(a)** **V** shows stereoisomerism.

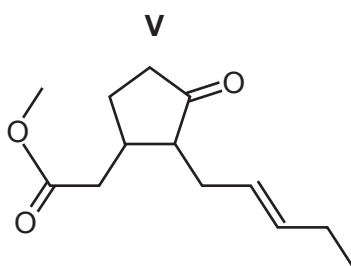


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

[1]

(ii) Deduce the number of stereoisomers of **V**. Explain your reasoning.

[2]

(iii) Deduce the molecular formula of **V**.

[1]

(iv) Name **all** the functional groups present in **V**.

[1]

34. M/J 24/P22/Q4/a,e

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

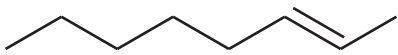
- (a) Draw the three-dimensional structures of the **two** stereoisomers of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$.
R can be used to represent $\text{CH}_3(\text{CH}_2)_5$.

[2]

- (e) (i) Both σ and π bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

	number of carbon atoms		
	sp hybridised	sp ² hybridised	sp ³ hybridised
E 			

[2]

- (ii) Describe the essential feature of an unbranched hydrocarbon that causes its molecules to show stereoisomerism. Explain how this feature leads to stereoisomerism.

[3]

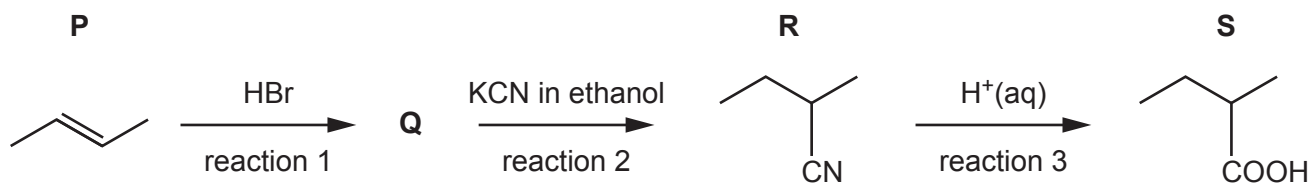
35. M/J 24/P23/Q4/e(iii)

- (iii) Give the systematic name for the organic product $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$.

[1]

36. MAR 24/P22/Q4/a

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

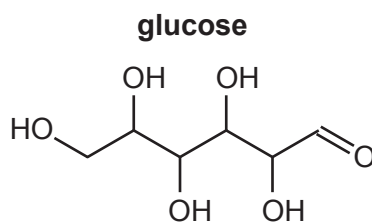
	P	Q	R	S
geometrical isomerism				
optical isomerism		✓		

[2]

37. O/N 23/P21/Q3/e(iii)

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

(iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

Define structural isomers.

[1]

38. O/N 23/P21/Q4/a(i,ii,iii)

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .



Fig. 4.1

(a) (i) Give the systematic name of **D**.

[1]

(ii) Explain why **C** and **D** do not show geometrical (*cis/trans*) isomerism.

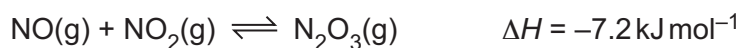
[1]

(iii) Draw the structure of a molecule that is a positional isomer of **C** and **D**.

[1]

39. O/N 23/P22/Q2/e(i)

2 NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

(e) NO₂ is a brown gas that can be used to form nitric acid.

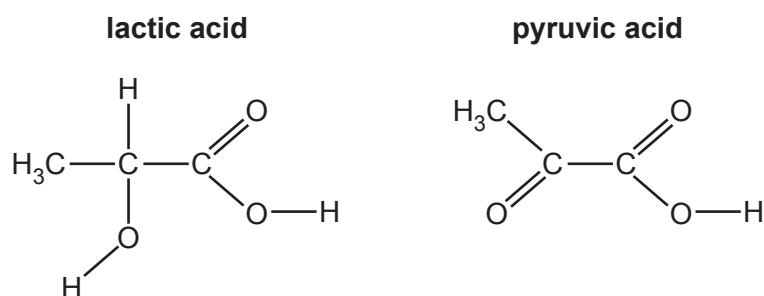
(i) NO₂ is a free radical.

Define free radical.

[1]

40. O/N 23/P22/Q4/a(i,ii)

- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

- (a) (i) Explain why lactic acid exists as optical isomers.

[1]

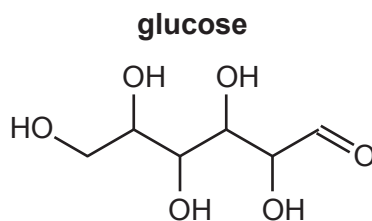
- (ii) Give the systematic name of lactic acid.

[1]

41. O/N 23/P23/Q3/e(iii)

- 3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

- (e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

Define structural isomers.

[1]

42. O/N 23/P23/Q4/a(i,ii,iii)

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .

**Fig. 4.1**

(a) (i) Give the systematic name of **D**.

[1]

(ii) Explain why **C** and **D** do not show geometrical (*cis/trans*) isomerism.

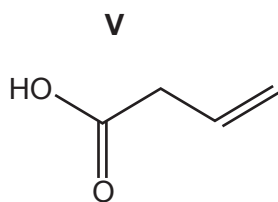
[1]

(iii) Draw the structure of a molecule that is a positional isomer of **C** and **D**.

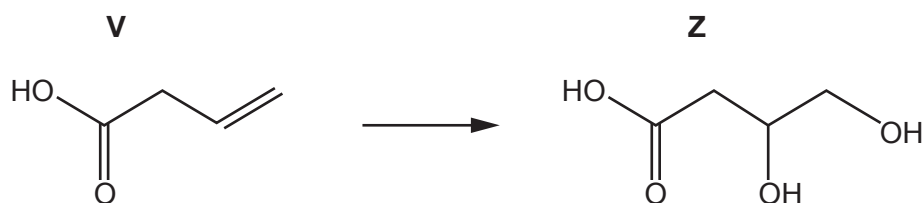
[1]

43. M/J 23/P21/Q4/b(iii)

4 **V** is a colourless liquid.

**Fig. 4.1**

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

**Fig. 4.2**

(iii) Complete Table 4.1 to show the number of sp^2 and sp^3 hybridised carbon atoms that are present in a molecule of **V**.

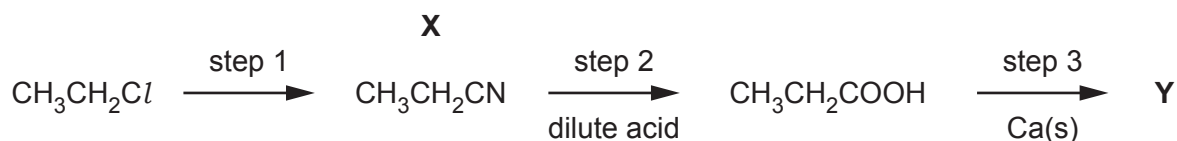
Table 4.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in V		

[2]

44. M/J 23/P22/Q3/a(ii),c

- 3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (ii) Give the systematic name of **X**.

[1]

- (c) Complete Table 3.1 to show the number of sigma bonds (σ) and pi bonds (π) present in a molecule of **X**.

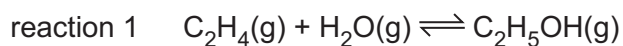
Table 3.1

type of bond	number of bonds in X
sigma (σ)	
pi (π)	

[2]

45. M/J 23/P22/Q4/d

- 4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (d) Describe the covalent bonds present between the carbon atoms in an ethene molecule by completing Table 4.2.

Table 4.2

	sigma (σ)	pi (π)
type of orbitals involved in bond		
how the orbitals overlap		

[2]

46. M/J 23/P22/Q5/a,c

5 (a) Describe structural isomerism.

[1]

(c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

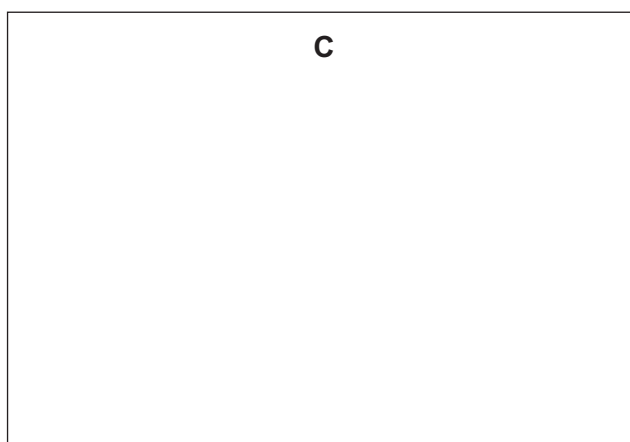
C shows only one type of stereoisomerism.

Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	C
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
Br ₂ (aq)	orange to colourless
alkaline I ₂ (aq)	yellow precipitate appears

(i) Draw the structure of **C** in the box.



[2]

(ii) Name the type of stereoisomerism shown by molecules of **C**.

[1]

47. M/J 23/P23/Q5/c

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

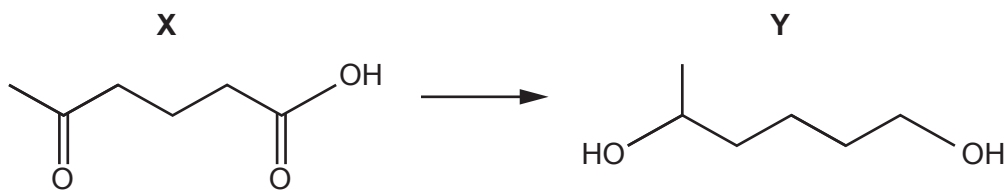


Fig. 5.1

(c) Complete Table 5.1 to show the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of **X**.

Table 5.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in X		

[2]

48. M/J 23/P23/Q6/c,d

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

(c) Molecules of **W**, $\text{CH}_2=\text{CHCN}$, do **not** show stereoisomerism.

(i) Describe stereoisomerism.

[1]

(ii) Describe the **two** essential features of an alkene molecule that cause it to show geometrical stereoisomerism.

[2]

(d) Molecules of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ exist as a pair of optical isomers.

Draw three-dimensional diagrams in the boxes to show the optical isomers of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

isomer 1	isomer 2
----------	----------

[1]

49. MAR 23/P22/Q3/a(i,ii,iii),c

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

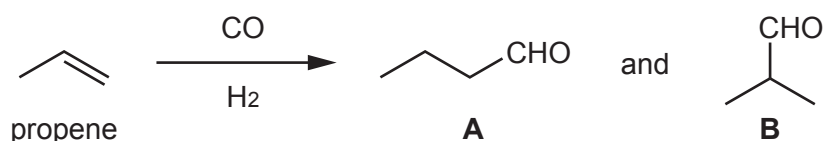


Fig. 3.1

- (a) (i)** Define addition reaction.

[1]

- (ii)** Aldehydes **A** and **B** are structural isomers.

State the type of structural isomerism shown by **A** and **B**.

[1]

- (iii)** Name **A**.

[1]

- (c)** When propene reacts with CO and an excess of H₂, an alkane and a mixture of alcohols are formed instead. The alcohols are isomers of each other.

Suggest the molecular formulae of the alkane and the alcohols that are formed under these conditions.

molecular formula of alkane

molecular formula of alcohols

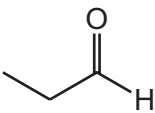
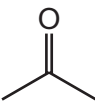
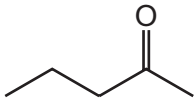
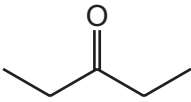
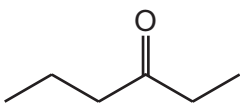
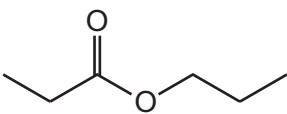
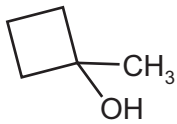
[2]

50. O/N 22/P22/Q3/a(ii,iii,iv)

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(a) Table 3.1 shows four pairs of organic compounds.

Table 3.1

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(ii) **A1** and **A2** are structural isomers.

Define structural isomers.

[1]

(iii) Give the systematic name of **B2**.

[1]

(iv) Deduce the molecular formula of **D1**.

[1]

51. M/J 22/P21/Q3/a

3 Liquids that contain molecules of **T** smell like lemons.

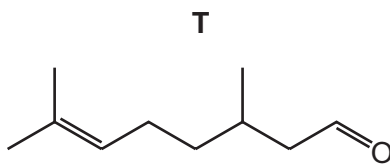


Fig. 3.1

(a) Molecules of **T** exist as a pair of stereoisomers.

Name the type of stereoisomerism shown by molecules of **T**. Explain your answer.

[2]

52. M/J 22/P21/Q4/c(iii)

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.

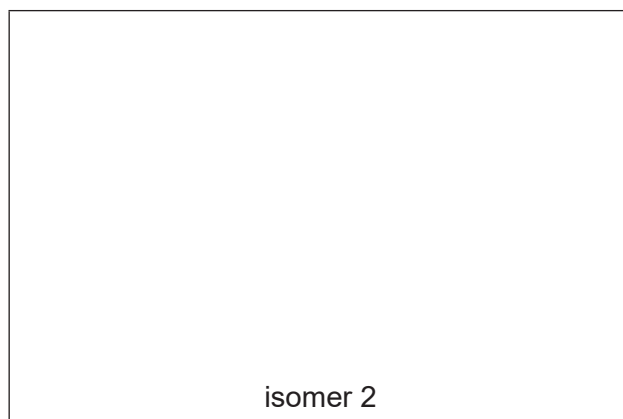
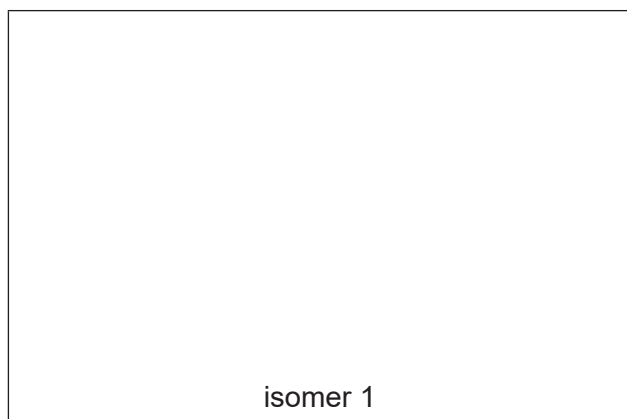
W, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.

Table 4.1

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(iii) **W** could be one of four structural isomers.

- Draw the skeletal formulae for two possible structural isomers of **W**.
- Describe the type of structural isomerism shown.



type of structural isomerism

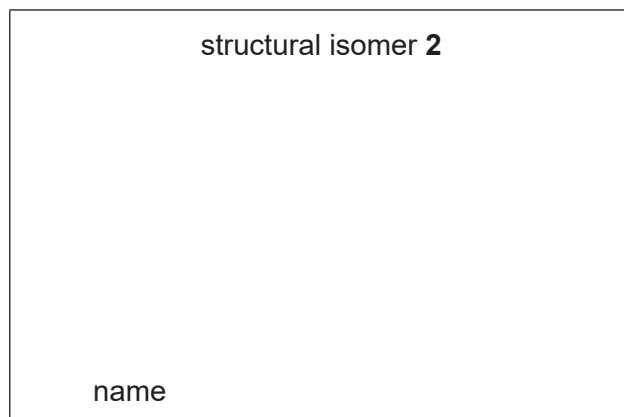
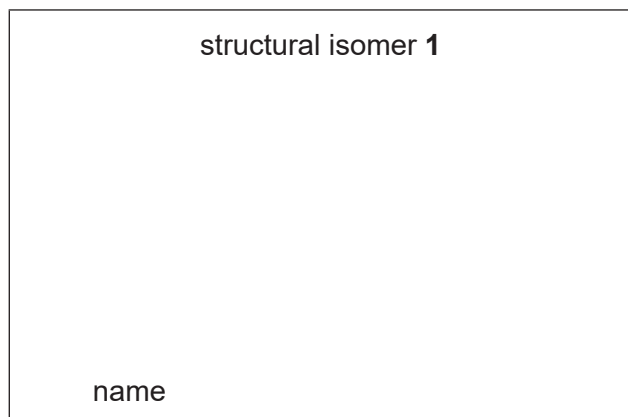
[3]

53. M/J 22/P22/Q4/a,b(i,ii)

- 4 (a)** Bromine reacts with butane in the presence of ultraviolet light to form bromobutane.

Two structural isomers with the molecular formula C_4H_9Br are produced during this reaction.

- (i)** Draw the two structural isomers and state the systematic name of each isomer.



[2]

- (ii)** Identify the type of structural isomerism shown in **(a)(i)**.

[1]

- (b)** Halothane is an anaesthetic.

halothane

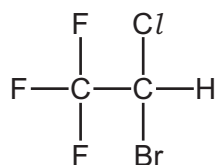


Fig. 4.1

- (i)** Identify the chiral centre in halothane and mark it with an asterisk (*). [1]

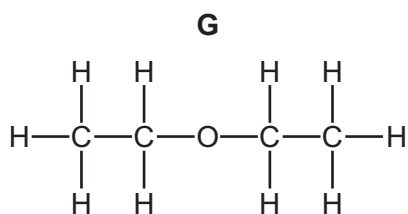
When halothane reacts in ultraviolet light, homolytic fission occurs and the C–Br bond is broken.

- (ii)** Construct an equation to show the homolytic fission of halothane, $CF_3CHBrCl$.

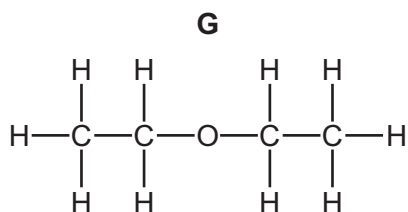
[1]

54. M/J 22/P23/Q3/b

- 3 G** belongs to a group of compounds called ethers.

**Fig. 3.1**

- (b) The functional group in **G** is an oxygen atom bonded to two carbon atoms.

**Fig. 3.3**

G, **H** and **J** are structural isomers with molecular formula $C_4H_{10}O$.

H and **J** are straight chain molecules.

Table 3.3 shows the boiling points and reactions of **G**, **H** and **J** when heated under reflux with excess acidified $K_2Cr_2O_7$.

Table 3.3

	G	H	J
boiling point/ $^{\circ}C$	35	117	98
heat under reflux with excess acidified $K_2Cr_2O_7$	remains orange	orange to green	orange to green

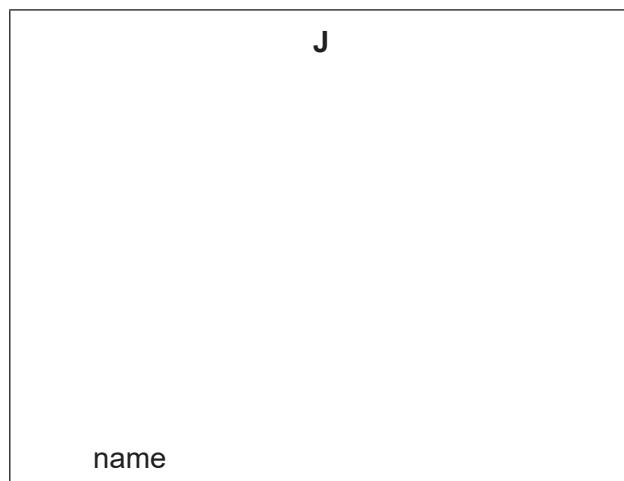
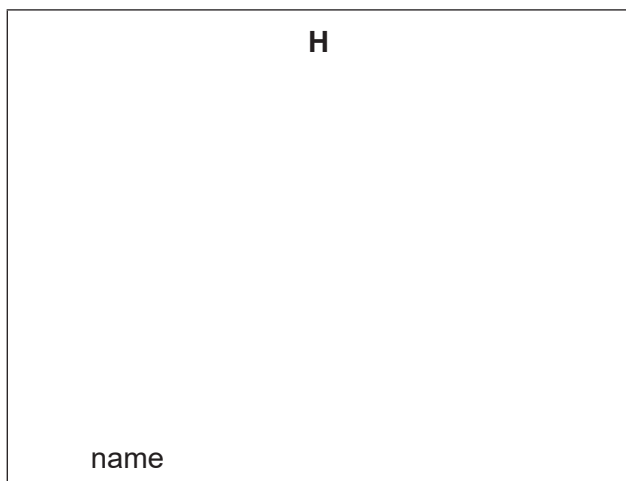
- (i) Identify the type of structural isomerism shown between **G** and **H** using the information in Table 3.3.

[1]

- (ii) Identify the type of structural isomerism shown between **H** and **J** using the information in Table 3.3.

[1]

(iii) Draw a possible structure for **H** and for **J**. State the systematic name for each structure.



[3]

55. MAR 22/P22/Q4/a(ii)

4 Compounds **J** and **K** are found in plant oils.

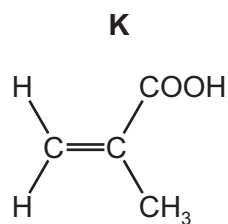
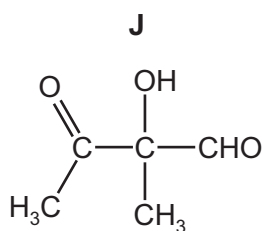


Fig. 4.1

(ii) **J** has **two** optical isomers.

Draw the three-dimensional structures of the **two** optical isomers of **J**.

.....

[2]

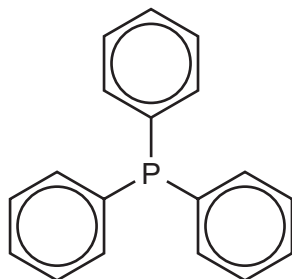
56. O/N 21/P21/Q3/c

also Unit 21 (2 marks), Unit 2 (1 mark); this unit 2 marks

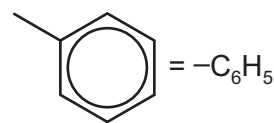
3 Phosphorus is a reactive Period 3 element.

(c) Triphenylphosphine is used in a type of reaction known as a *Wittig reaction*.

triphenylphosphine



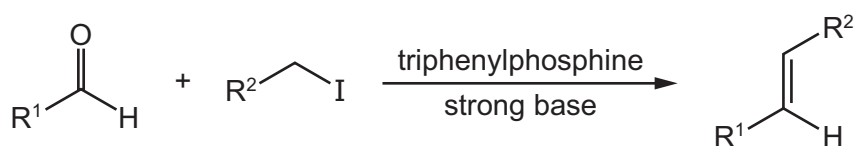
where



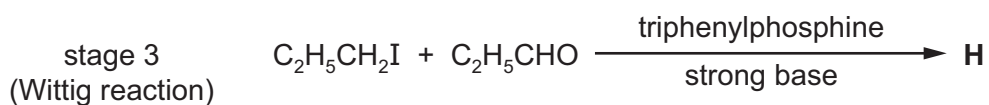
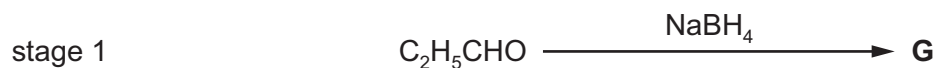
(i) Give the empirical formula of triphenylphosphine.

[1]

In a Wittig reaction, an aldehyde reacts with a halogenoalkane to form an alkene. The conversion is shown in the following unbalanced equation.



Compound **H** can be made from propanal, $\text{C}_2\text{H}_5\text{CHO}$. Stage 3 in the reaction scheme is a Wittig reaction.



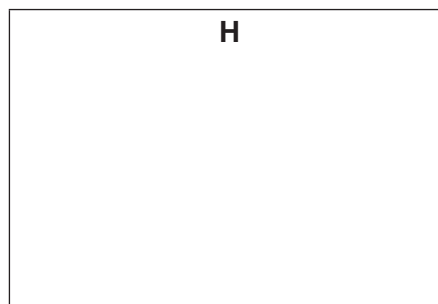
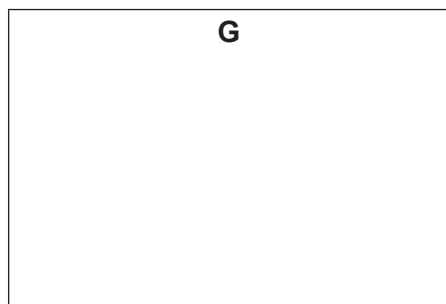
(ii) State the types of reaction that occur in stages 1 and 2.

stage 1

stage 2

[2]

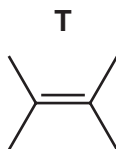
(iii) Draw the structures of **G** and **H** in the boxes provided.



[2]

57. O/N 21/P22/Q3/a,b,c,e(iv)

3 Compound **T** is an isomer of C_6H_{12} .



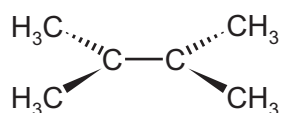
(a) Name **T**.

[1]

(b) Draw the skeletal formula of a structural isomer of **T** that shows *cis-trans* (geometrical) isomerism.

[1]

(c) Each carbon atom in **T** forms a sigma (σ) bond to at least one other carbon atom, as shown.



(i) On the diagram, draw the orbitals that represent the pi (π) bond that is also present in **T**.

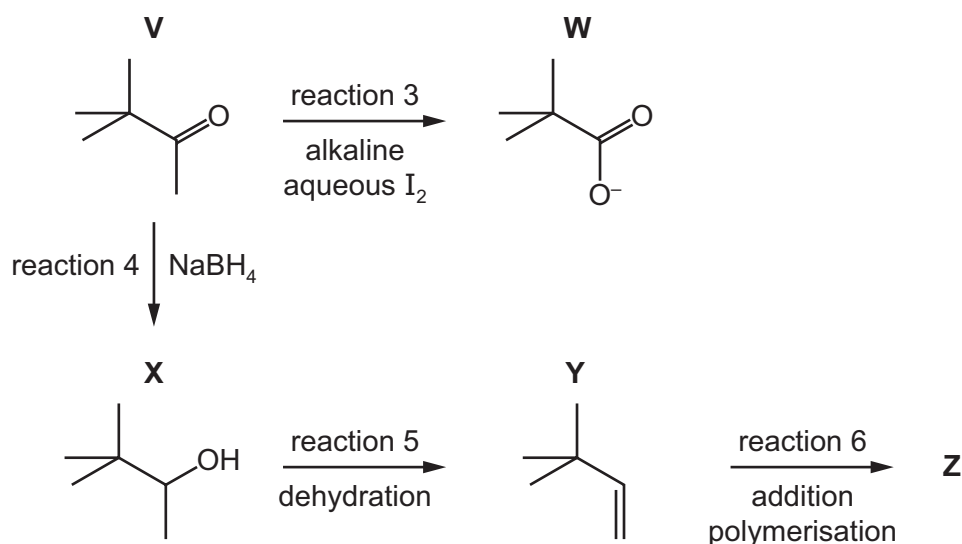
[1]

(ii) State the hybridisation of the two carbon atoms between which the pi (π) bond forms.

[1]

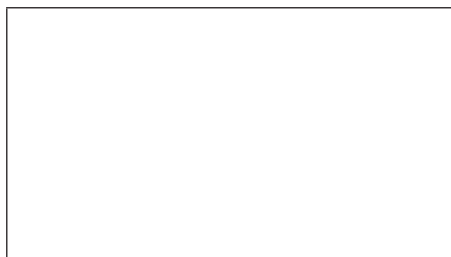
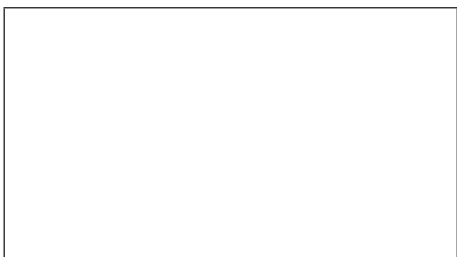
(e) **V** is used in a wide range of organic reactions.

Some reactions of **V** are shown.



(iv) **X** is a mixture of two optical isomers.

Draw the two optical isomers in the boxes provided.



[2]

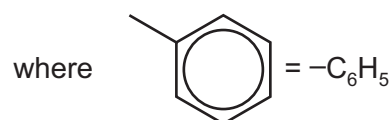
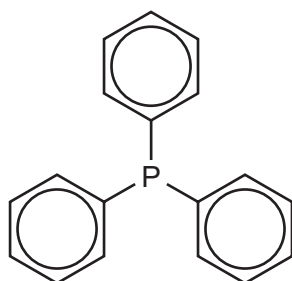
58. O/N 21/P23/Q3/c

also Unit 21 (2 marks), Unit 2 (1 mark); this unit 2 marks

3 Phosphorus is a reactive Period 3 element.

(c) Triphenylphosphine is used in a type of reaction known as a *Wittig reaction*.

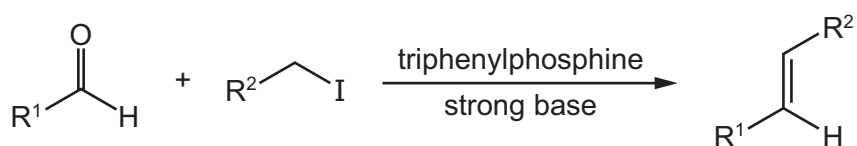
triphenylphosphine



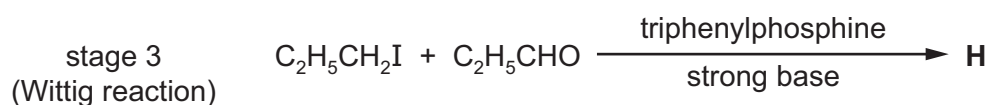
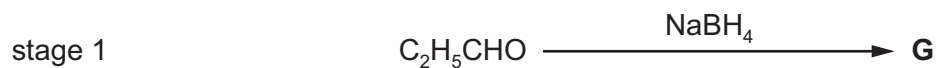
(i) Give the empirical formula of triphenylphosphine.

[1]

In a Wittig reaction, an aldehyde reacts with a halogenoalkane to form an alkene. The conversion is shown in the following unbalanced equation.



Compound **H** can be made from propanal, $\text{C}_2\text{H}_5\text{CHO}$. Stage 3 in the reaction scheme is a Wittig reaction.



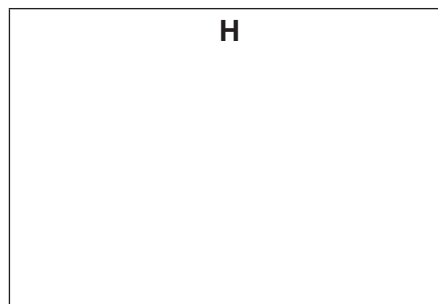
(ii) State the types of reaction that occur in stages 1 and 2.

stage 1

stage 2

[2]

(iii) Draw the structures of **G** and **H** in the boxes provided.



[2]

59. M/J 21/P21/Q5/a(i)

5 (a) Naphtha is a mixture which contains only hydrocarbon molecules.

(i) What is meant by the term *hydrocarbon*?

[1]

60. M/J 21/P21/Q6/c

6 Propene, C_3H_6 , reacts with H_2O in the presence of an acid catalyst to form an alcohol with molecular formula $\text{C}_3\text{H}_8\text{O}$.

(c) Complete the table to show the numbers of sigma (σ) bonds and pi (π) bonds present in propene, C_3H_6 , and $\text{C}_3\text{H}_8\text{O}$.

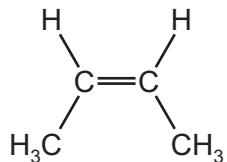
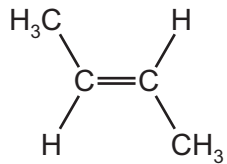
	σ	π
C_3H_6		
$\text{C}_3\text{H}_8\text{O}$		

[2]

61. M/J 21/P22/Q4/a

- 4 (a) The table shows the structural formulae of four compounds, **A**, **B**, **C** and **D**, with molecular formula C_4H_8 .

(i) Complete the table by giving the systematic name of **A**, **B**, **C** and **D**.

	structural formula	name
A	$CH_3CH_2CH=CH_2$	
B		
C		
D	$CH_2=C(CH_3)_2$	

[4]

(ii) Explain what is meant by *stereoisomerism*.

[1]

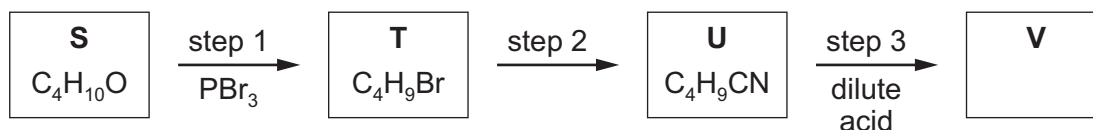
62. M/J 21/P22/Q5/a,b(i)

- 5 **S** is a secondary alcohol with molecular formula $C_4H_{10}O$.

(a) Draw the displayed formula of **S**.

[1]

(b) **S** is converted to **V** in a three-step reaction sequence.



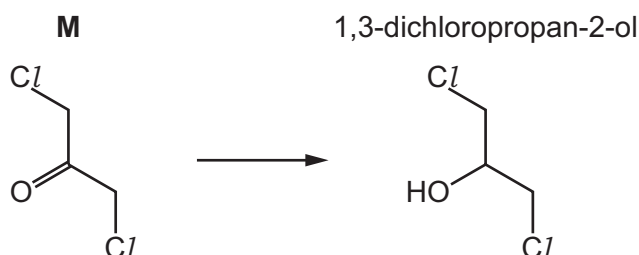
In step 1, the secondary alcohol **S** reacts with PBr_3 to produce **T**, which has molecular formula C_4H_9Br .

(i) Give the systematic name of **T**.

[1]

63. M/J 21/P23/Q4/a(i)

- 4 (a) 1,3-dichloropropan-2-ol can be made by reacting **M**.

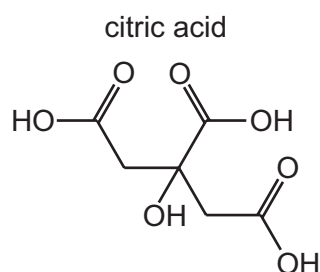


- (i) Give the systematic name of **M**.

[1]

64. M/J 21/P23/Q5/c,d

- 5 Compound **X** contains the same functional groups as citric acid.



The table describes some of the similarities and differences between citric acid and compound **X**.

	citric acid	X
chiral centre	no	yes
reaction with Na	fizzing	fizzing
reaction with $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	remains orange	orange to green

- (c) Compound **X** is one of a pair of stereoisomers.

Stereoisomerism occurs when a molecule has at least one of two key features.

State the two key features that give rise to stereoisomerism.

1

2

[2]

- (d) A structural isomer of compound **X** does not fizz when added to sodium.

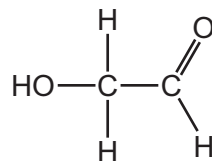
Explain what is meant by *structural isomer*.

[1]

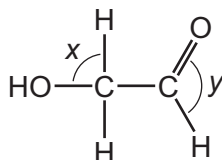
65. MAR 21/P22/Q4/a

- 4 Hydroxyethanal, HOCH_2CHO , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



- (a) Predict the bond angles labelled x and y in the diagram of hydroxyethanal.



$x =$

°

$y =$

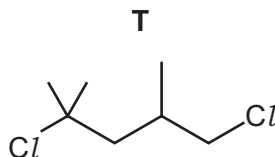
°

[2]

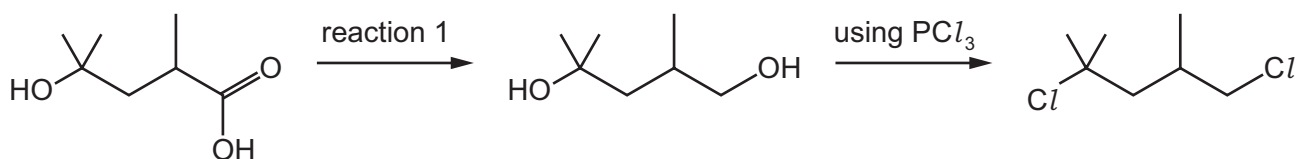
66. O/N 20/P22/Q3/c(ii)

- 3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

- (c) PCl_3 is used to convert alcohols to chloroalkanes, such as compound **T**.

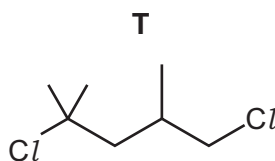


A possible synthesis of **T** is shown.



- (ii) **T** exhibits optical isomerism.

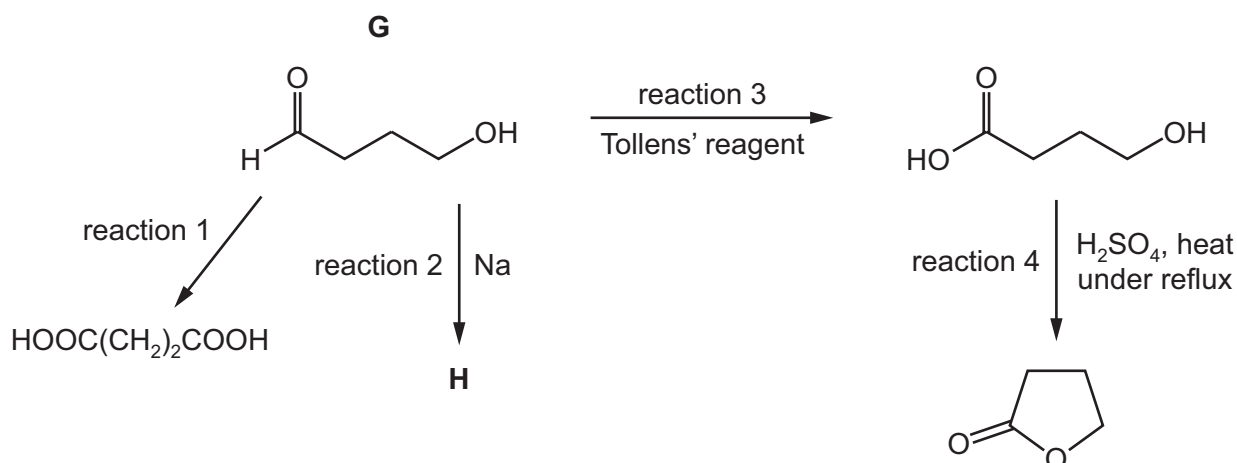
Explain what is meant by the term *optical isomer* and circle any atom(s) in **T** that give rise to optical isomerism.



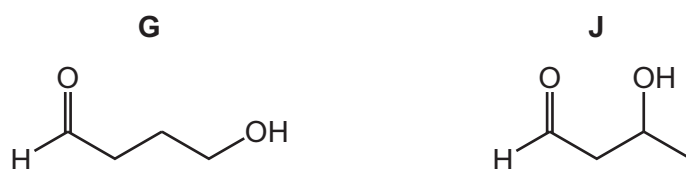
[2]

67. O/N 20/P22/Q4/b(i)

4 Some reactions of compound **G** are shown.



(b) **G** and **J** are structural isomers of each other.



(i) Name the type of structural isomerism shown by **G** and **J**.

[1]

68. M/J 20/P21/Q5/b(ii)

(b) $Cl(g)$ can be made from $Cl_2(g)$.

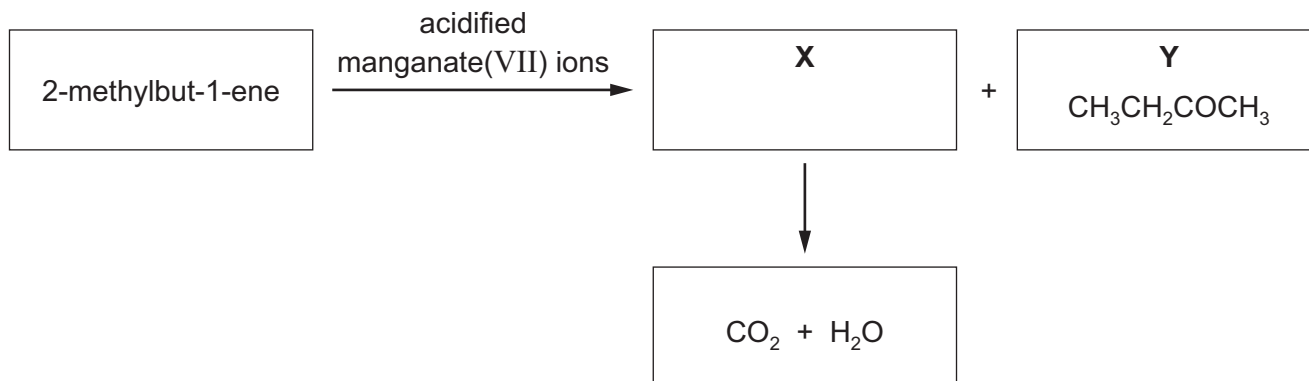
(ii) Name this process.

[1]

69. M/J 20/P21/Q6/a,g(i)

- 6** 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.



- (a)** Draw the skeletal formula of 2-methylbut-1-ene.

[1]

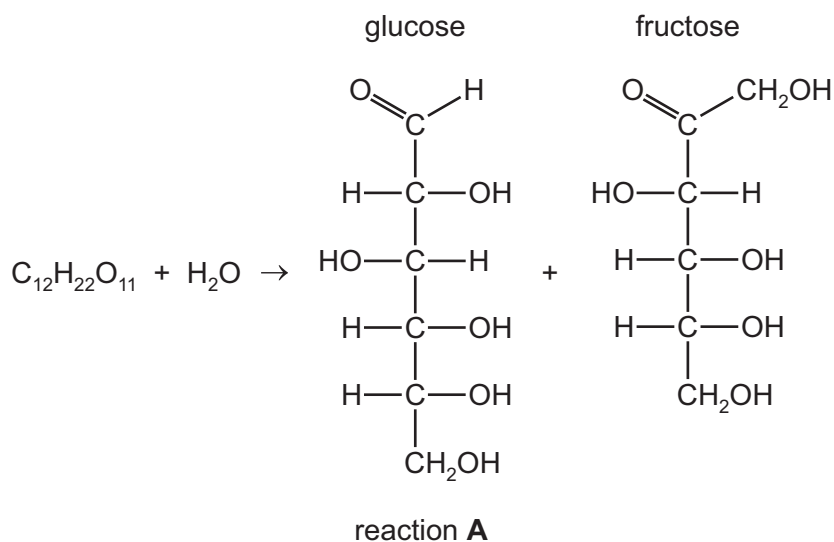
- (g)** Organic compound **W** is an ester which is a structural isomer of propanoic acid.

- (i)** State the molecular formula of **W**.

[1]

70. M/J 20/P22/Q3/a,b

- 3** Sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, reacts with water to form glucose and fructose in reaction **A**.



- (a)** Suggest a name for this type of reaction.

[1]

- (b)** Explain in detail, why glucose and fructose are a pair of structural isomers. Your answer should refer specifically to these two molecules.

[2]

71. M/J 20/P22/Q4/c(iii,iv)

- (c) Two different halogenoalkanes, **P** and **R**, both with the molecular formula C_4H_9Cl , are separately dissolved in ethanol and heated under reflux with sodium hydroxide.

The major organic product of each of these reactions is methylpropene.

- (iii) Draw the skeletal formula of methylpropene.

[1]

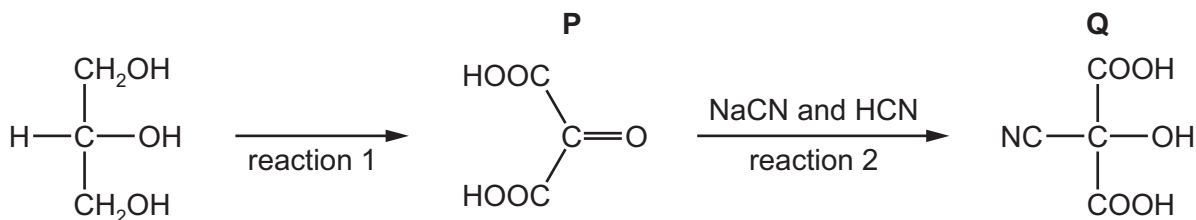
- (iv) Give the names of **P** and **R**.

[2]

72. MAR 20/P22/Q3/a(iv),c(i),d(i)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(a) A series of reactions starting from glycerol is shown.



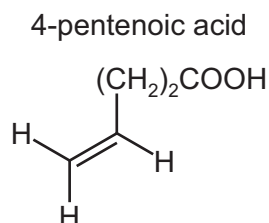
(iv) **Q** does **not** show optical isomerism.

Explain why.

[1]

(c) Fats are compounds made from glycerol and unsaturated carboxylic acids.

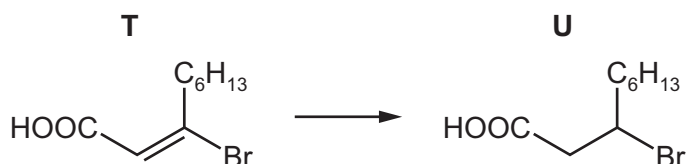
4-pentenoic acid is an example of an unsaturated carboxylic acid.



(i) Give the molecular formula of 4-pentenoic acid.

[1]

(d) A reaction of another unsaturated carboxylic acid, **T**, is shown.



(i) **T** is one of a pair of geometrical (*cis-trans*) isomers.

Draw the other geometrical isomer of **T** and explain why the molecules exhibit this form of isomerism.

[3]

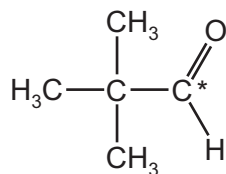
73. O/N 19/P21/Q4/a,c(iii,iv)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(a) (i) Give the name of $(\text{CH}_3)_3\text{CCHO}$.

[1]

(ii) State the hybridisation of the carbon atom labelled with an asterisk, *.



[1]

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

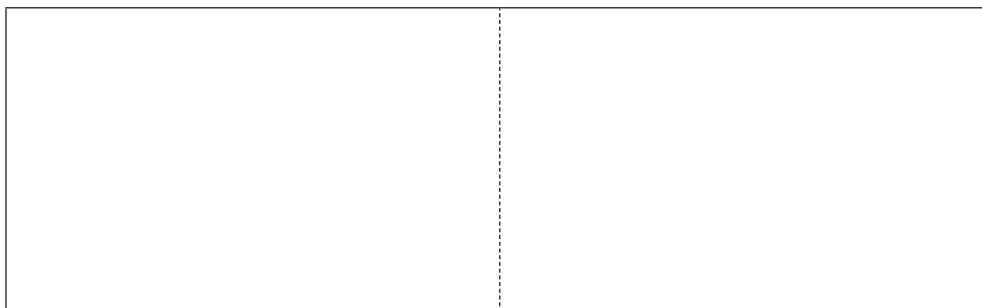
compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

(iii) **Y** has a chiral centre and exists as a pair of optical isomers.

State what is meant by the term *chiral centre*.

[1]

(iv) Draw the optical isomers of **Y** using the conventional three-dimensional representation.

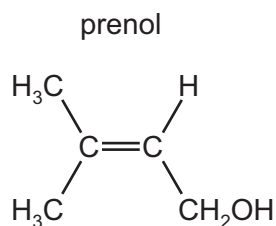


[2]

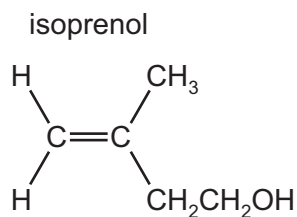
Z, $\text{C}_5\text{H}_{10}\text{O}$, has a branched carbon chain. It shows geometrical isomerism.

74. O/N 19/P22/Q4/d(i)

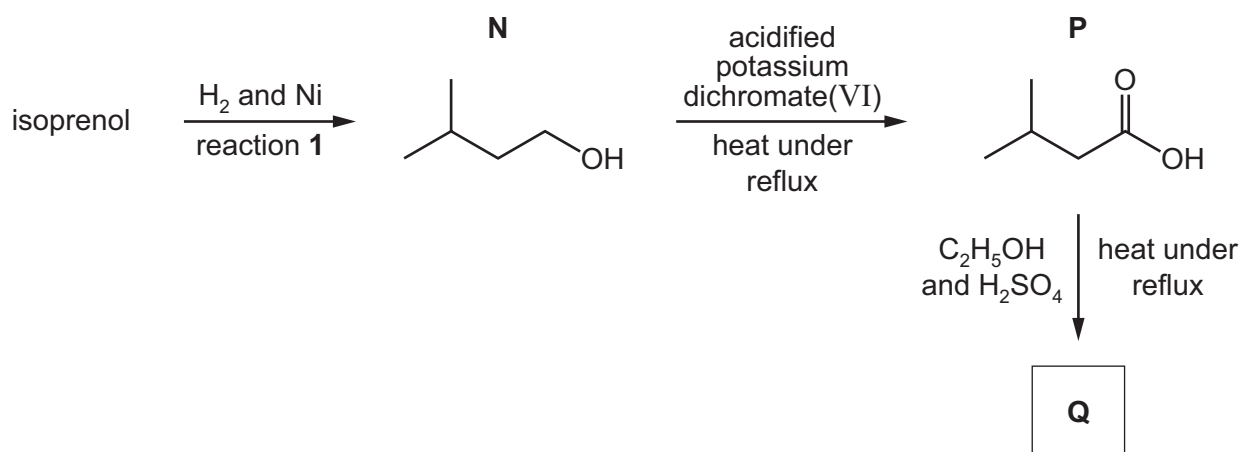
- 4 Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (d) Isoprenol is a structural isomer of prenol.



The series of reactions shows how isoprenol can be used to form **Q**, a sweet-smelling liquid.



- (i) Give the name of **N**.

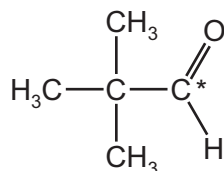
75. O/N 19/P23/Q4/a,c(iii,iv)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(a) (i) Give the name of $(\text{CH}_3)_3\text{CCHO}$.

[1]

(ii) State the hybridisation of the carbon atom labelled with an asterisk, *.



[1]

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

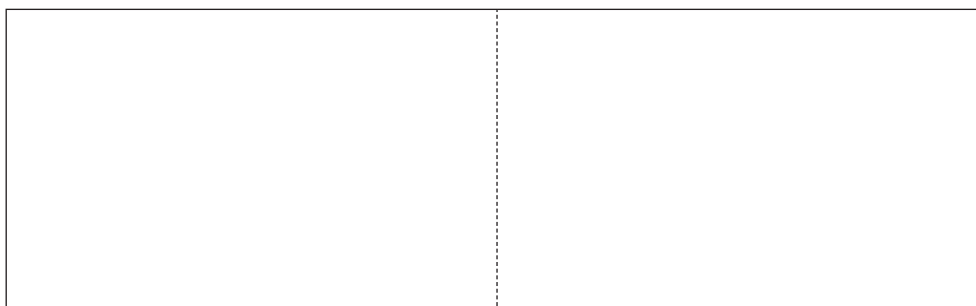
compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

(iii) **Y** has a chiral centre and exists as a pair of optical isomers.

State what is meant by the term *chiral centre*.

[1]

(iv) Draw the optical isomers of **Y** using the conventional three-dimensional representation.

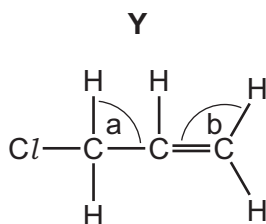


[2]

Z, $\text{C}_5\text{H}_{10}\text{O}$, has a branched carbon chain. It shows geometrical isomerism.

76. M/J 19/P21/Q4/a,b,c(i)

4 The structure of compound **Y** is shown.



(a) Give the systematic name for **Y**.

[1]

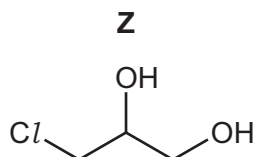
(b) Predict the values for the bond angles **a** and **b** shown in the diagram.

a

b

[2]

(c) When **Y** reacts with cold, dilute, acidified manganate(VII) ions, compound **Z** is produced.



(i) State the molecular formula of **Z**.

[1]

77. M/J 19/P21/Q5/b

5 Ethanal reacts with a mixture of HCN and NaCN to make 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

The reaction mechanism is nucleophilic addition.

(b) $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ exists as a pair of stereoisomers.

(i) Name the type of stereoisomerism shown by $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

[1]

(ii) Draw three-dimensional diagrams of this pair of stereoisomers.

Indicate with an asterisk (*) the chiral centre on one of the structures drawn.

[3]

78. M/J 19/P22/Q1/a

1 Methylpropane, $(\text{CH}_3)_2\text{CHCH}_3$, is an isomer of butane, $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$.

(a) (i) Explain why methylpropane and butane are a pair of isomers.

[2]

(ii) Identify the type of isomerism shown by methylpropane and butane.

[1]

79. M/J 19/P22/Q4/d(i,ii,v)

4 There are many different types of aliphatic and aromatic hydrocarbons.

(d) The table compares the reactivity of alkanes and alkenes with chlorine.

	alkanes	alkenes
name of the type of reaction with chlorine	substitution	addition and substitution
name of the type of reacting species	free radical	electrophile and free radical

(i) During the first stage in the substitution reaction chlorine forms chlorine free radicals.

Explain what is meant by the term *free radical*.

[1]

(ii) Name and explain the type of bond breaking which occurs to form chlorine free radicals.

[2]

(v) Carbon atoms can form σ and π bonds within hydrocarbon molecules.

Explain the following statement with reference to σ and π bonds.

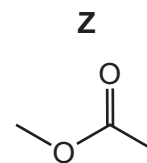
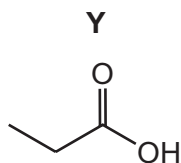
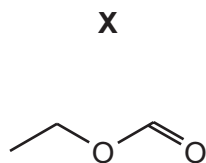
Alkenes react with electrophiles but alkanes do not.

[2]

80. M/J 19/P22/Q5/c(i)

5 Many naturally occurring esters are used as flavourings in food.

(c) Compounds **X**, **Y** and **Z** are shown. They all have the same molecular formula.



(i) Deduce the molecular formula of **X**, **Y** and **Z**.

[1]

81. M/J 19/P23/Q6/a(iii)

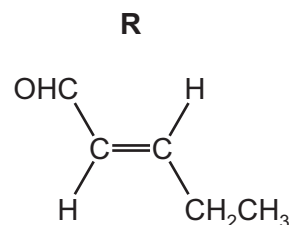
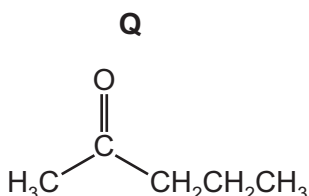
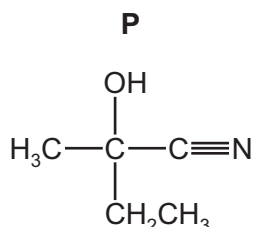
6 (a) Three alkenes, **X**, **Y** and **Z**, have the same molecular formula.

(iii) Deduce the molecular formula of **X**, **Y** and **Z**.

[1]

82. MAR 19/P22/Q3/c

3 **P**, **Q** and **R** all contain five carbon atoms.

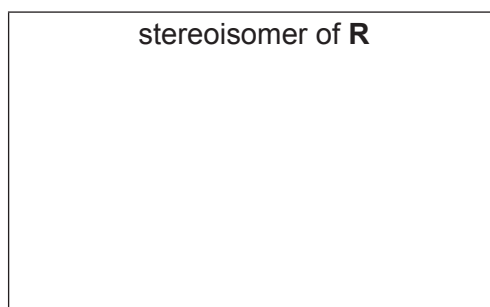


A student carries out several tests to distinguish between **P**, **Q** and **R**.

(c) **R** exists as a pair of stereoisomers.

Identify the type of stereoisomerism shown by **R** and draw the structure of the other stereoisomer.

type of stereoisomerism

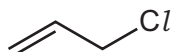


[2]

83. MAR 19/P22/Q4/a

4 Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride

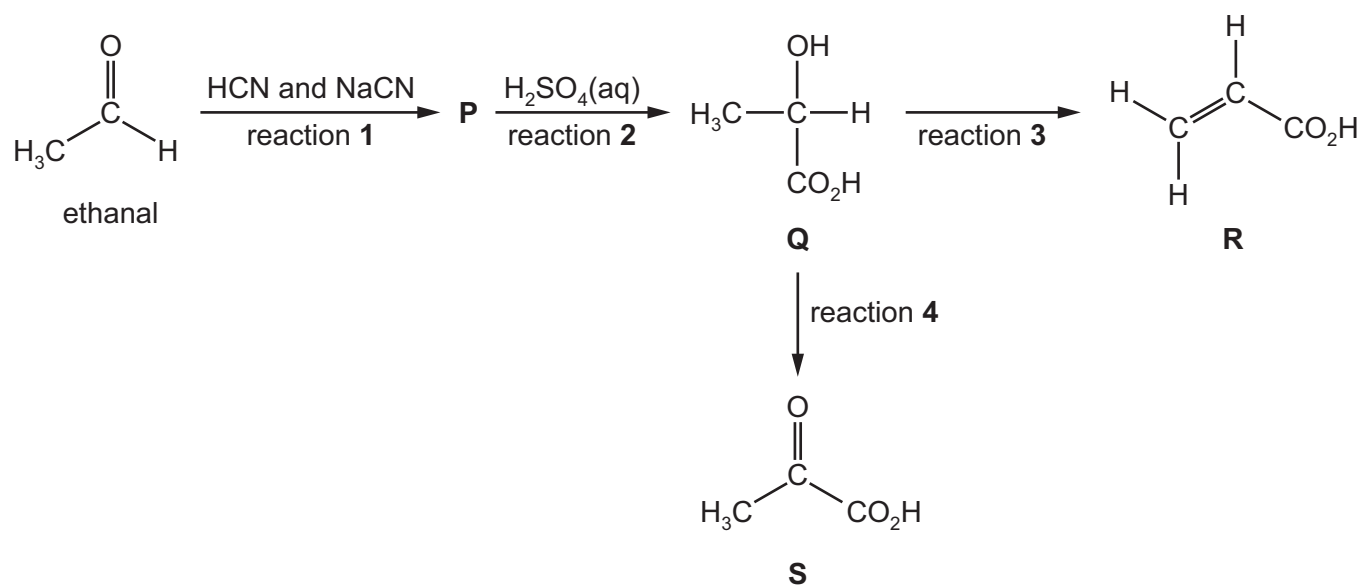


(a) Give the systematic name of allyl chloride.

[1]

84. O/N 18/P22/Q4/b

4 The diagram shows a reaction sequence starting from ethanal.



(b) Compound **Q** is formed as a mixture of two optical isomers.

(i) Explain what is meant by the term *optical isomers*.

[1]

(ii) Draw the **two** optical isomers of **Q**, showing clearly their three-dimensional structures.

.....

[2]

85. M/J 18/P22/Q4/a(ii),b,d(ii,iii)

4 **W** is $\text{CH}_3\text{COCH}_2\text{CH}_3$.

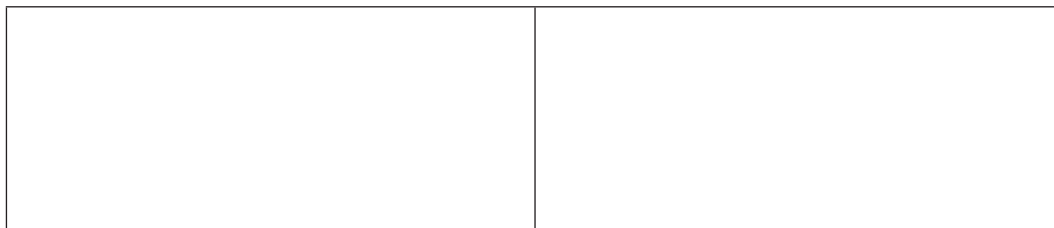
(a) The reaction between **W** and alkaline aqueous iodine produces a yellow precipitate.

(ii) Give the name of **W**.

[1]

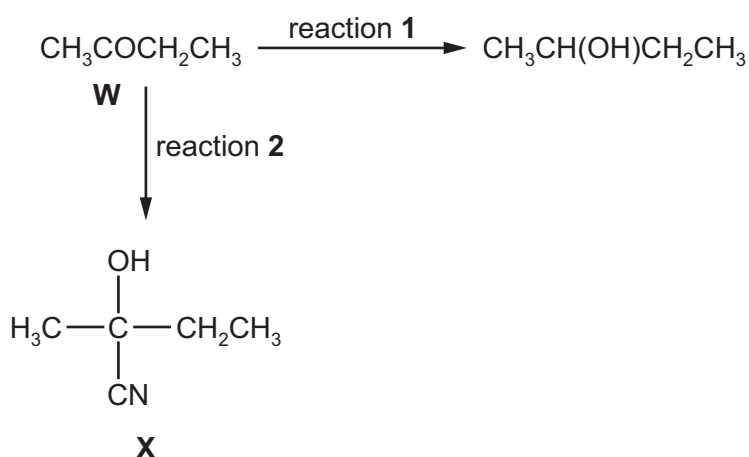
(b) There are two structural isomers of **W** that are also carbonyl compounds.

Draw the structures of these two isomers of **W**.



[2]

Two reactions of **W** are shown.



(d) Reaction 2 is carried out by adding a mixture of HCN and NaCN to **W**.
The product, **X**, is formed as a mixture of two isomers.

(ii) State the name of the type of isomerism shown by **X**.

[1]

(iii) Explain fully why **X** shows this type of isomerism.

[2]

86. M/J 18/P23/Q2/a,b(i,iii)

- 2** One reason for the wide variety of organic compounds is isomerism, either structural isomerism or stereoisomerism.

(a) (i) Explain the meaning of the term *structural isomerism*.

[2]

(ii) Explain the meaning of the term *stereoisomerism*.

[2]

(b) Pent-1-ene, $\text{CH}_2=\text{CH}(\text{CH}_2)_2\text{CH}_3$, does not show stereoisomerism.

(i) Give **two** reasons why pent-1-ene does **not** show stereoisomerism.

reason 1

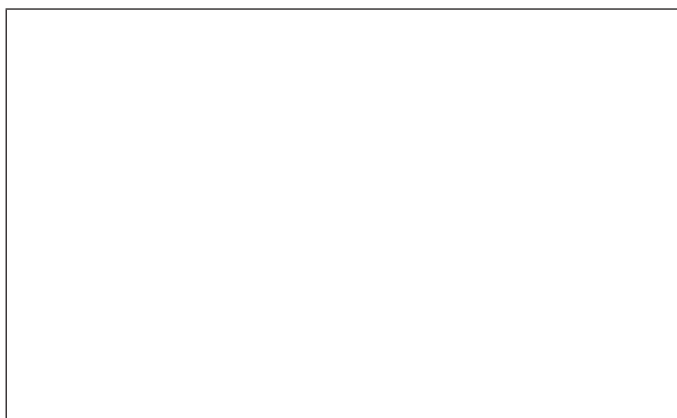
reason 2

[2]

(iii) A different structural isomer of pent-1-ene shows geometrical isomerism.

Draw the structure of **one** of the two geometrical isomers with the formula C_5H_{10} .

Give the full name of this isomer.

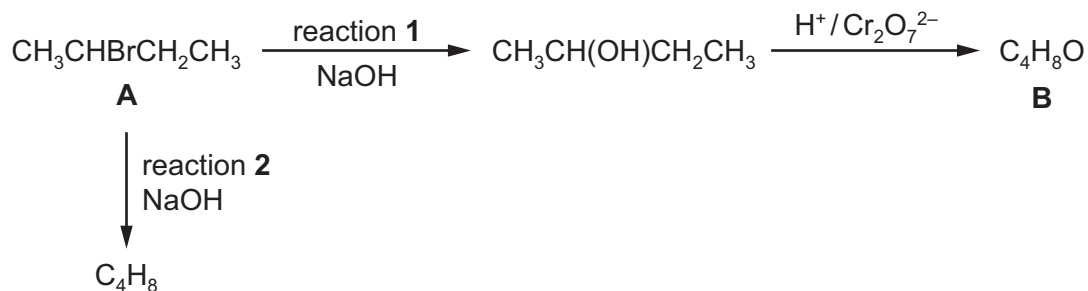


[2]

87. M/J 18/P23/Q4/a(i),b

4 **A** is $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$.

(a) Some reactions of **A** are shown.



(i) Name **A**.

[1]

(b) There are three structural isomers of **A**.

Draw the structures of these three isomers of **A**.

--	--	--

[2]

88. MAR 18/P22/Q2/b(i)

2 Carbon and silicon are elements in Group 14.

(b) C_{60} forms hydrocarbons with similar chemical properties to those of alkenes. One such hydrocarbon is $\text{C}_{60}\text{H}_{18}$.

(i) State what is meant by the term *hydrocarbon*.

[1]

89. MAR 18/P22/Q3/b(ii)

3 Calcium and its compounds have a large variety of applications.

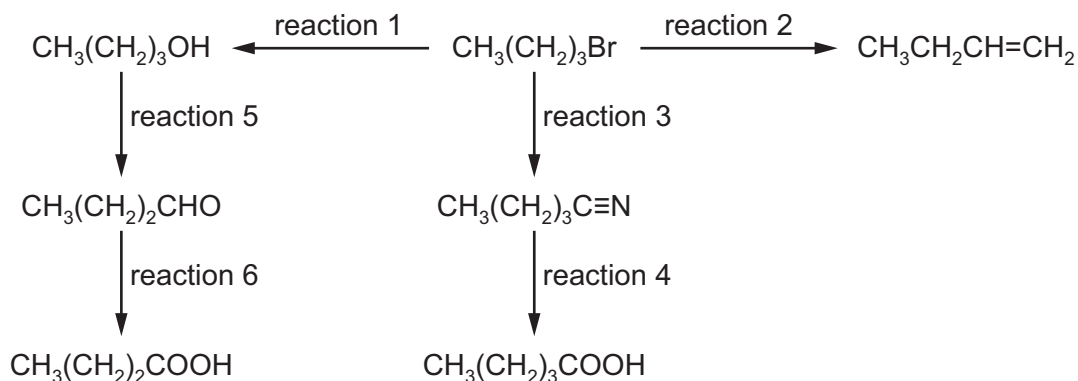
(b) Calcium ethanedioate is formed when calcium reacts with ethanedioic acid, $(\text{CO}_2\text{H})_2$. The compound contains one cation and one anion.

(ii) Draw the displayed formula of the anion present in calcium ethanedioate.

[2]

90. O/N 17/P21/Q3/c(i),d(i,ii,iv,v)

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



(c) 2-bromo-2-methylpropane is a tertiary halogenoalkane that is a structural isomer of 1-bromobutane.

(i) Define the term *structural isomer* and name the three different types of structural isomerism.

definition

types of structural isomerism

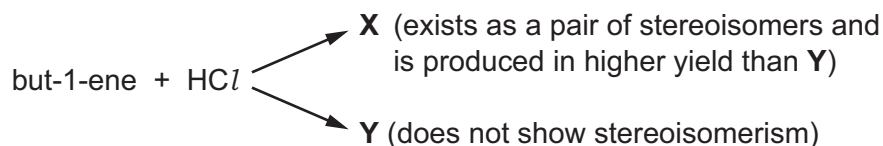
1

2

3

[4]

(d) The product of reaction 2, but-1-ene, does **not** show stereoisomerism. However, but-1-ene reacts with HCl to form a mixture of structural isomers **X** and **Y**.



(i) Explain the meaning of the term *stereoisomers*.

[2]

(ii) Give **two** reasons why but-1-ene does **not** show stereoisomerism.

[2]

(iv) Name the type of stereoisomerism shown by **X**.

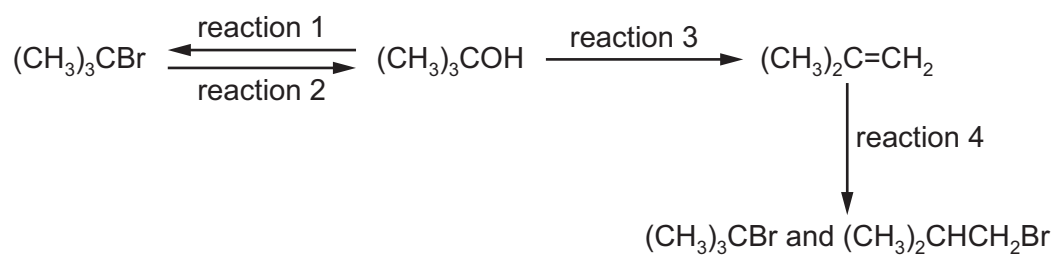
[1]

(v) Use the conventional representation to draw the two stereoisomers of **X**.

[2]

91. O/N 17/P22/Q4/c(i),d(i)

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



(c) 1-bromobutane is a structural isomer of the product of reaction 1.

(i) Define the term *structural isomer* and name the three different types of structural isomerism.

definition

types of structural isomerism

1

2

3

[4]

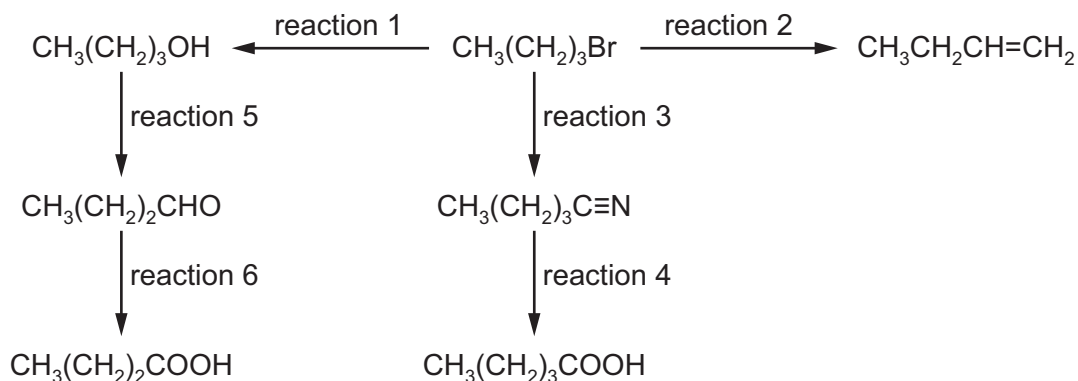
(d) The product of reaction 3, methylpropene, does **not** show stereoisomerism.

(i) Give **two** reasons why methylpropene does **not** show stereoisomerism.

[2]

92. O/N 17/P23/Q3/c(i),d(i,ii,iv,v)

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



(c) 2-bromo-2-methylpropane is a tertiary halogenoalkane that is a structural isomer of 1-bromobutane.

(i) Define the term *structural isomer* and name the three different types of structural isomerism.

definition

types of structural isomerism

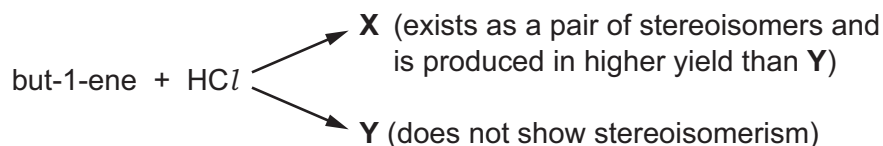
1

2

3

[4]

(d) The product of reaction 2, but-1-ene, does **not** show stereoisomerism. However, but-1-ene reacts with HCl to form a mixture of structural isomers **X** and **Y**.



(i) Explain the meaning of the term *stereoisomers*.

[2]

(ii) Give **two** reasons why but-1-ene does **not** show stereoisomerism.

[2]

(iv) Name the type of stereoisomerism shown by **X**.

[1]

(v) Use the conventional representation to draw the two stereoisomers of **X**.

[2]

93. M/J 17/P21/Q1/b(iv)

- 1** Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds.

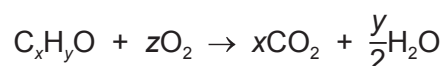
(b) **T** is an alcohol, C_xH_yO . A gaseous sample of **T** occupied a volume of 20 cm^3 at 120°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess). The final volume, measured under the same conditions as the gaseous sample, was 250 cm^3 .

Under these conditions, all water present is vaporised. Removal of the water vapour from the gaseous mixture decreased the volume to 170 cm^3 .

Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 110 cm^3 .

The equation for the complete combustion of **T** can be represented as shown.



- (iv)** Give the skeletal formulae for two possible structures of **T**.

Name each alcohol.



[2]

94. M/J 17/P21/Q4/a,d(i)

4 **P**, **Q** and **R** all have the molecular formula $\text{C}_3\text{H}_6\text{O}$. They are all structural isomers of each other.

P and **Q** each contain an oxygen atom bonded directly to a carbon atom that is sp^2 hybridised.
R contains an oxygen atom bonded directly to a carbon atom that is sp^3 hybridised.

(a) (i) Explain the meaning of the term *structural isomers*.

[2]

(ii) Explain how sp^2 and sp^3 hybridisation can occur in carbon atoms.

sp^2 hybridisation

sp^3 hybridisation

[2]

(iii) State the bond angles normally associated with each type of hybridisation in carbon atoms.

sp^2

sp^3

[2]

(d) **P** and **Q** each react with hydrogen cyanide to form a single product.

The product formed from **P** exists as a pair of optical isomers.

The product formed from **Q** does not exhibit optical isomerism.

(i) Explain the meaning of the term *optical isomers*.

[2]

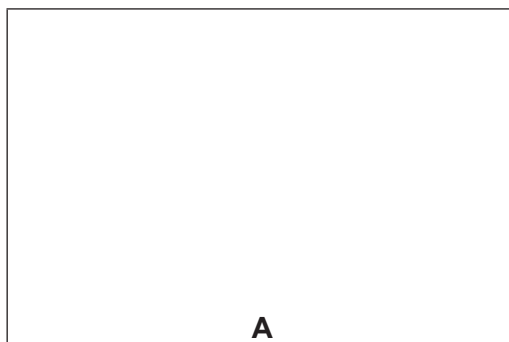
95. M/J 17/P22/Q4/a(i,iii,iv),c(iii)

4 (a) The hydrocarbons **A**, C_4H_{10} , and **B**, C_4H_8 , are both unbranched.

A does **not** decolourise bromine.

B decolourises bromine and shows geometrical isomerism.

(i) Draw the skeletal formula of **A**.



[1]

(iii) Give the structural formula of **B**.

[1]

(iv) Explain why **B** shows geometrical isomerism.

[2]

(c) The isomers **F** and **G**, $C_5H_{10}O$, both form an orange precipitate when reacted with 2,4-DNPH.

F is unbranched and reacts with alkaline aqueous iodine to produce a yellow precipitate.

G does not react with alkaline aqueous iodine. It contains a chiral centre and produces a silver mirror when warmed with Tollens' reagent.

(iii) Explain the meaning of the term *chiral centre*.

[1]

96. M/J 17/P23/Q1/a,b(iv)

- 1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds. Combustion data cannot distinguish between different structural isomers.

(a) Define the term *structural isomers*

[2]

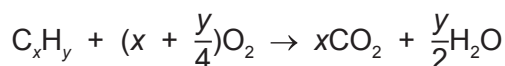
- (b) **P** is a hydrocarbon, C_xH_y . A gaseous sample of **P** occupied a volume of 25 cm^3 at 37°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess).

The final volume, measured under the same conditions as the gaseous sample (so that the water produced is liquid and its volume can be ignored), was 150 cm^3 .

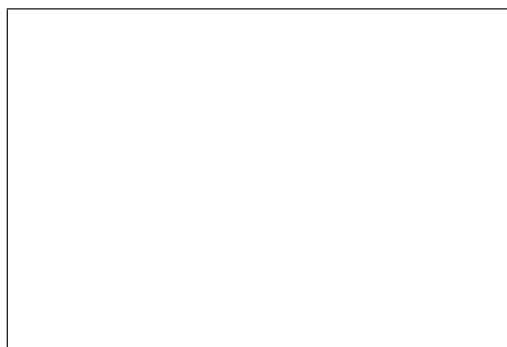
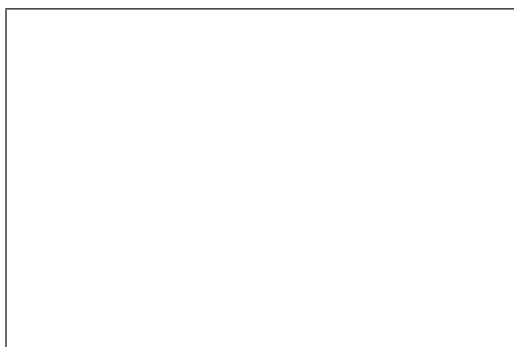
Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 50 cm^3 .

The equation for the complete combustion of **P** can be represented as shown.



- (iv) **P** is unbranched.

Give the skeletal formulae for two possible structures of **P** that are positional isomers of each other.

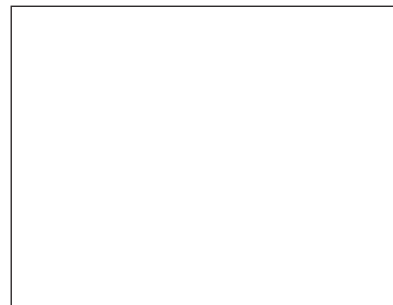
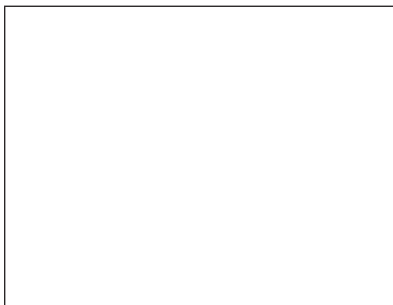
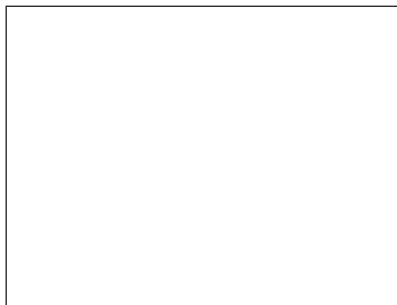


[2]

97. M/J 17/P23/Q4/b,c(i),d(i,ii,iii),e(i)

- 4 **A**, **B** and **C** all have the formula C_4H_8 . They all decolourise bromine and are structural isomers of each other.

(b) Draw the structures of these three **structural** isomers.



[1]

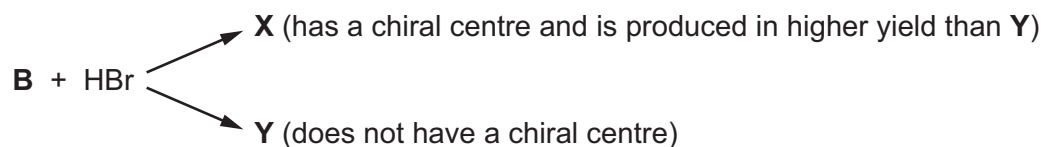
(c) Only **A** shows geometrical isomerism.

(i) Explain the meaning of the term *geometrical isomerism*.

[2]

(d) **B** does **not** show geometrical isomerism.

B reacts with HBr to form a mixture of two structural isomers, **X** and **Y**.



(i) State the meaning of the term *chiral centre*.

[1]

(ii) Name **B**.

[1]

(iii) **X** exists as a pair of optical isomers.

Draw these isomers using the conventional three-dimensional representation.



[2]

(e) **C** does **not** show geometrical isomerism.

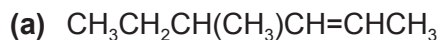
C reacts with HBr to form a mixture of two structural isomers, neither of which has a chiral centre.

(i) Name **C**.

[1]

98. O/N 16/P21/Q4/a,b(i),c(i)

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.



(i) name [1]

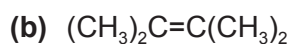
(ii) This compound shows stereoisomerism.

Define *stereoisomerism*.

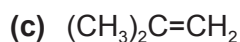
[1]

(iii) State and explain how many stereoisomers of this structure there are.

[4]



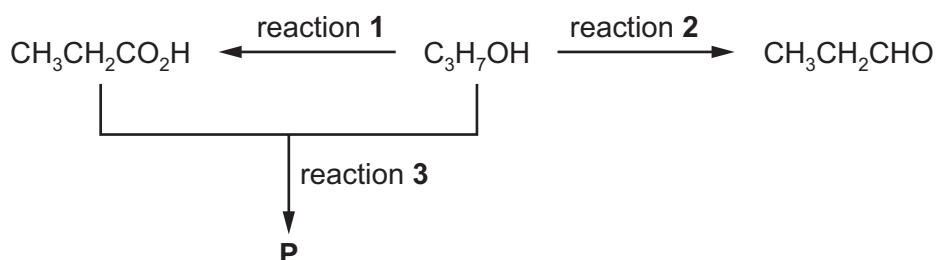
(i) name [1]



(i) name [1]

99. O/N 16/P21/Q5/a

- 5** A sequence of reactions is shown starting with an alcohol, $\text{C}_3\text{H}_7\text{OH}$.

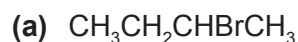


(a) Draw the skeletal formula of the alcohol $\text{C}_3\text{H}_7\text{OH}$.

[1]

100. O/N 16/P22/Q4/a(ii,iii)

- 4** In each section of this question the structural formula of an organic compound is shown. For each compound answer the questions about it.



- (ii)** This compound shows stereoisomerism.

Draw the **two** stereoisomers in the conventional way.

.....

[2]

- (iii)** Give the structures of **three** other structural isomers of $\text{C}_4\text{H}_9\text{Br}$.



[3]

101. O/N 16/P22/Q5/a(ii)

- 5** In each section of this question choose the answer or answers from the options listed.

- (a)** Six particles are listed.

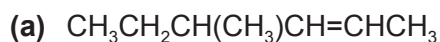


- (ii)** Identify the **two** particles produced by the heterolytic fission of a bond in chloromethane.

[1]

102. O/N 16/P23/Q4/a,b(i),c(i)

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.



(i) name [1]

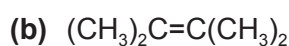
(ii) This compound shows stereoisomerism.

Define *stereoisomerism*.

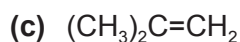
[1]

(iii) State and explain how many stereoisomers of this structure there are.

[4]



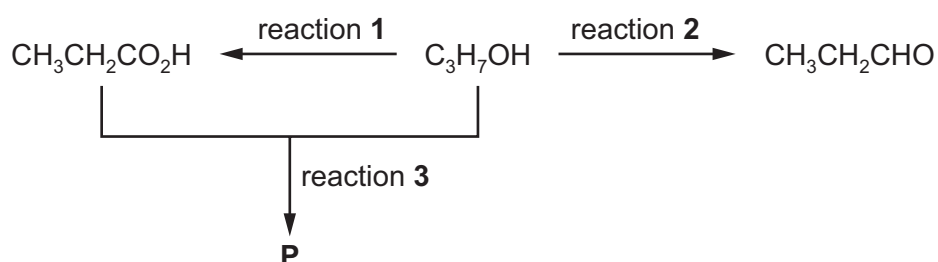
(i) name [1]



(i) name [1]

103. O/N 16/P23/Q5/a

- 5** A sequence of reactions is shown starting with an alcohol, $\text{C}_3\text{H}_7\text{OH}$.



(a) Draw the skeletal formula of the alcohol $\text{C}_3\text{H}_7\text{OH}$.

[1]

104. M/J 16/P21/Q4/a,b,c,d

4 This question is about molecules with molecular formula C_4H_8 .

(a) Give the structures of a pair of **positional** isomers with the formula C_4H_8 .



[1]

(b) Give the structures of a pair of **chain** isomers with the formula C_4H_8 , that do **not** exhibit stereoisomerism.



[1]

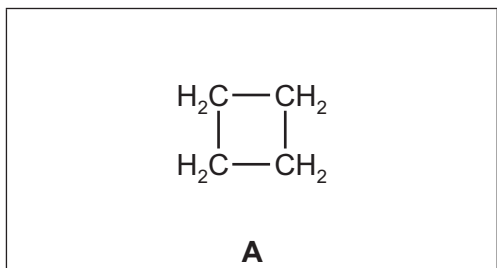
(c) Give the structures and full names of a pair of **stereoisomers** with the formula C_4H_8 .



[2]

(d) The structure of a molecule, **A**, of formula C_4H_8 is shown.

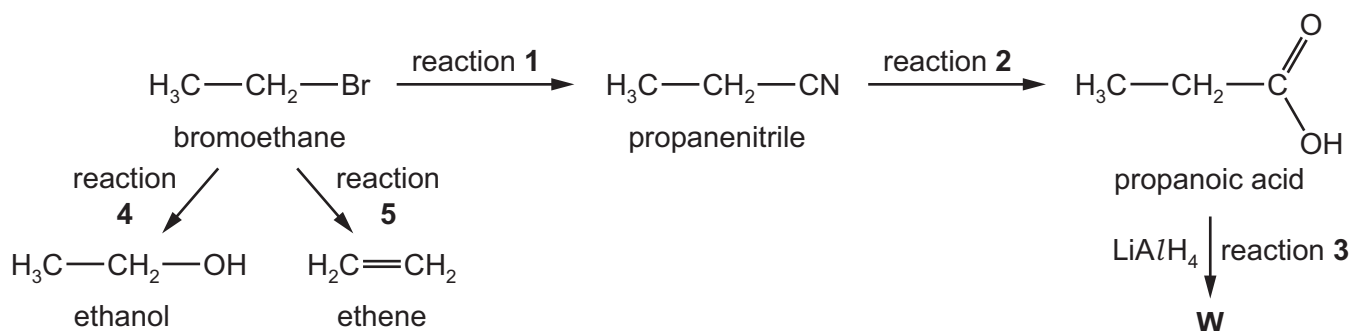
Draw a functional group isomer of molecule **A** in box **B**. Explain how molecules **A** and **B** could be distinguished by a chemical test.



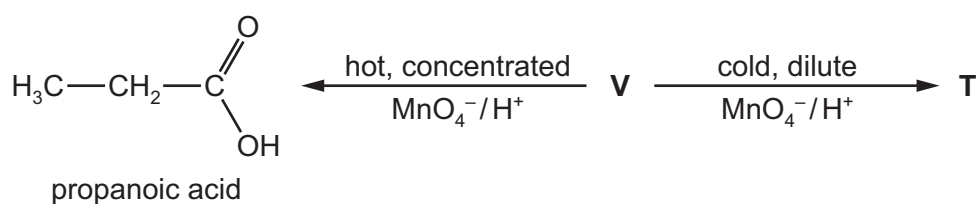
[3]

105. M/J 16/P21/Q5/e(ii)

5 A reaction sequence is shown.



(e) **V** reacts with acidified manganate(VII) ions in two different ways depending on the conditions, as shown in the reaction sequence below.



V decolourises bromine water.

When the acidified manganate(VII) is hot and concentrated, propanoic acid is the only organic product.

When the acidified manganate(VII) is cold and dilute, the organic product is **T** which has two chiral centres.

(ii) Identify the types of stereoisomerism shown by **V** and **T**.

V

T

[2]

106. M/J 16/P22/Q4/a,b

also Unit 18 (2 marks); this unit 4 marks

4 This question is about molecules with molecular formula $C_4H_8O_2$.

- (a)** Give the structural formulae of the pair of **chain** isomers with the formula $C_4H_8O_2$ that are carboxylic acids.



[2]

- (b) (i)** Give the structural formulae of a pair of **positional** isomers with the formula $C_4H_8O_2$ that are esters.



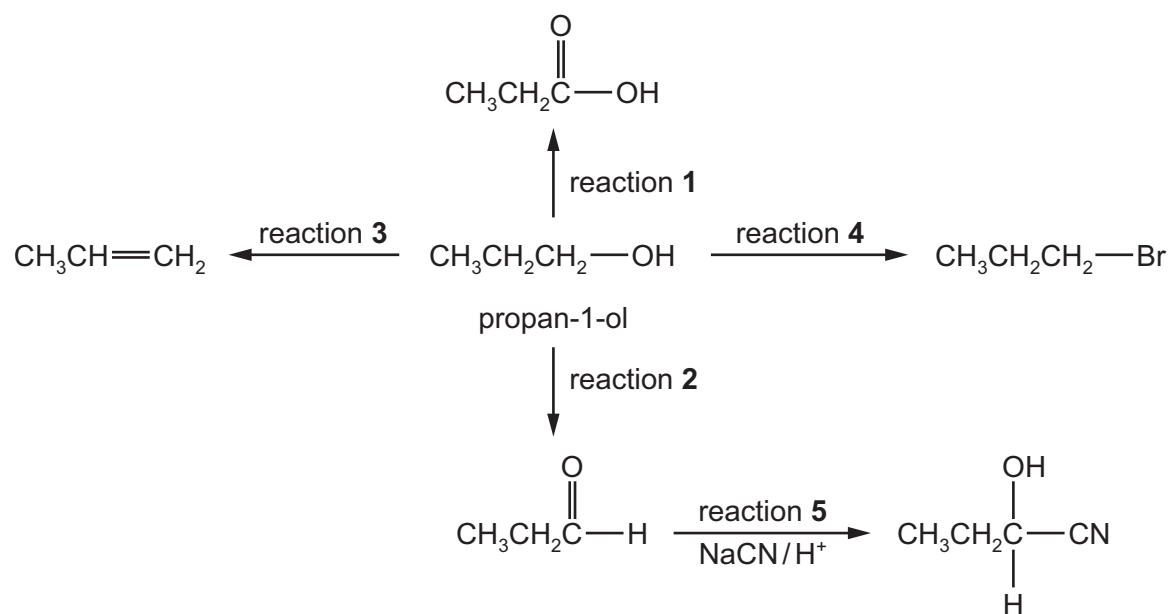
[2]

- (ii)** Give the reagents and conditions needed to produce one of your esters in **(i)**.

[2]

107. M/J 16/P22/Q5/c(ii)

5 A reaction sequence based on propan-1-ol is shown.



(ii) Draw the two stereoisomers in the conventional way.

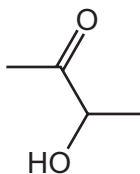
.....

[2]

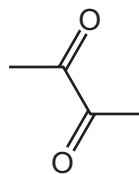
108. M/J 16/P23/Q4/a,d

also Unit 16 (1 mark); this unit 6 marks

- 4** Acetoin, $\text{CH}_3\text{COCH}(\text{OH})\text{CH}_3$, and diacetyl, $\text{CH}_3\text{COCOCH}_3$, are two of the compounds that give butter its characteristic flavour. Their skeletal formulae are shown.



acetoin



diacetyl

- (a)** Give the systematic name for acetoin.

[1]

- (d)** If a sample of acetoin is reacted with concentrated sulfuric acid, a single product is formed that does **not** exhibit stereoisomerism.

However, if a sample of acetoin is reacted with HBr, a mixture of a pair of stereoisomers is produced.

- (i)** Give the structural formula of the product of the reaction of acetoin with concentrated sulfuric acid.

[1]

- (ii)** Explain why the product in **(i)** does **not** exhibit stereoisomerism.

[2]

- (iii)** Explain why the product of reaction of acetoin with HBr **does** exhibit stereoisomerism.

[1]

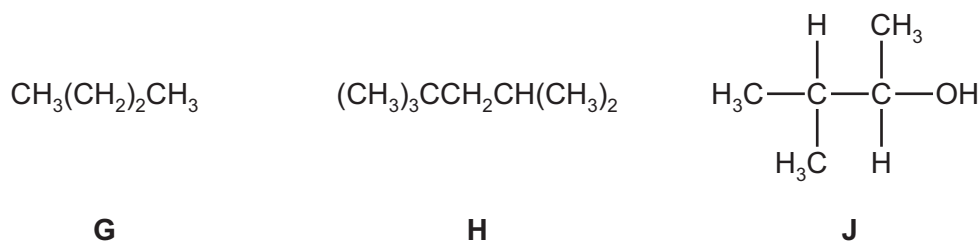
- (iv)** Draw the two stereoisomers from **(iii)** using the conventional representation.

.....

[2]

109. MAR 16/P22/Q4/a

4 The following compounds were all found to be components of a sample of petrol.



(a) (i) Give the **molecular** formula of compound **G**.

[1]

(ii) Give the **empirical** formula of compound **H**.

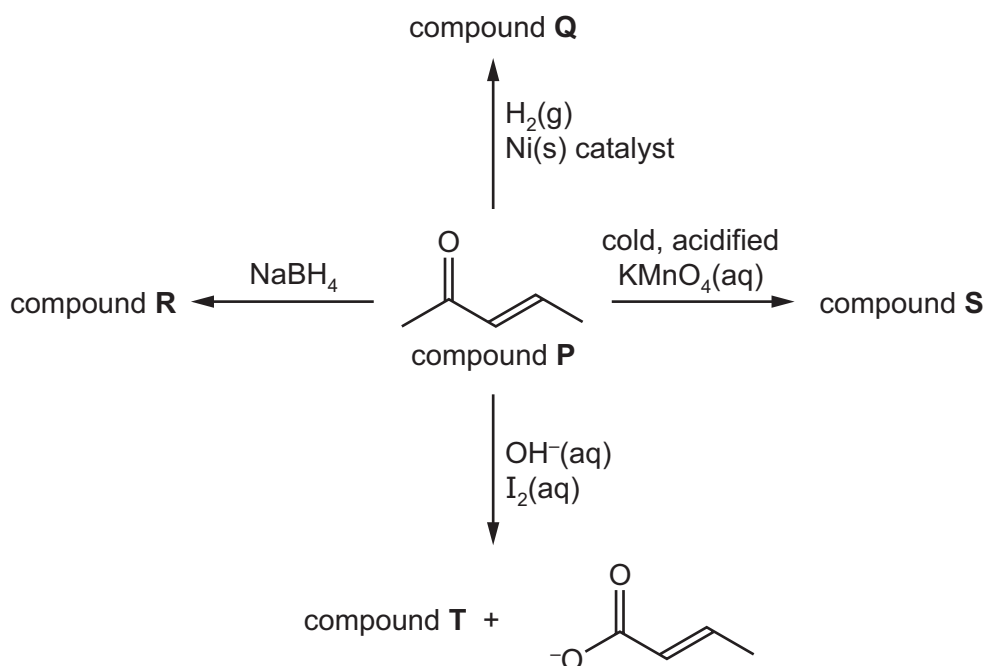
[1]

(iii) Draw the **skeletal** formula of compound **J**.

[1]

110. MAR 16/P22/Q5/a(ii)

5 Some reactions of compound **P**, $\text{C}_5\text{H}_8\text{O}$, are shown.



(ii) Give the systematic name of compound **P**.

[1]

111. O/N 15/P21/Q3/a

3 Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(a) There are nine structural isomers with the formula C_7H_{16} , only two of which contain chiral centres.

(i) Explain the meanings of the terms *structural isomers* and *chiral*.

structural isomers

chiral

[2]

(ii) Give the structures and names of the two structural isomers of C_7H_{16} which contain a chiral centre.

[4]

112. O/N 15/P22/Q4/a(ii,iii,v,vi)

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

(a) 2-bromobutane reacts in two different ways with sodium hydroxide depending on the conditions.

When warmed with aqueous sodium hydroxide, 2-bromobutane produces an alcohol that exists as a pair of optical isomers.

(ii) Explain why the alcohol produced exists as a pair of optical isomers.

[1]

(iii) Draw the three-dimensional structure of the two optical isomers of the alcohol produced in **(ii)**.

.....

[2]

Heating 2-bromobutane with ethanolic sodium hydroxide produces a mixture of three alkenes, two of which are a pair of geometrical isomers.

(v) Draw and name the structures of the pair of geometrical isomers formed by reaction of 2-bromobutane with ethanolic sodium hydroxide.

name

name

[2]

(vi) Name the third alkene produced by reaction of 2-bromobutane with ethanolic sodium hydroxide and explain why it does **not** show geometrical isomerism.

[2]

113. O/N 15/P23/Q3/a

- 3** Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(a) There are nine structural isomers with the formula C_7H_{16} , only two of which contain chiral centres.

(i) Explain the meanings of the terms *structural isomers* and *chiral*.

structural isomers

chiral

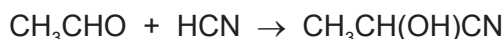
[2]

(ii) Give the structures and names of the two structural isomers of C_7H_{16} which contain a chiral centre.

[4]

114. M/J 15/P21/Q3/b(i)

- 3** Ethanal reacts with hydrogen cyanide, in the presence of a small amount of NaCN, as shown.



(b) The product of this reaction shows stereoisomerism as it contains a chiral centre. This reaction produces an equimolar mixture of two optical isomers.

(i) Explain the meanings of the terms *stereoisomerism* and *chiral centre*.

stereoisomerism

chiral centre

[2]

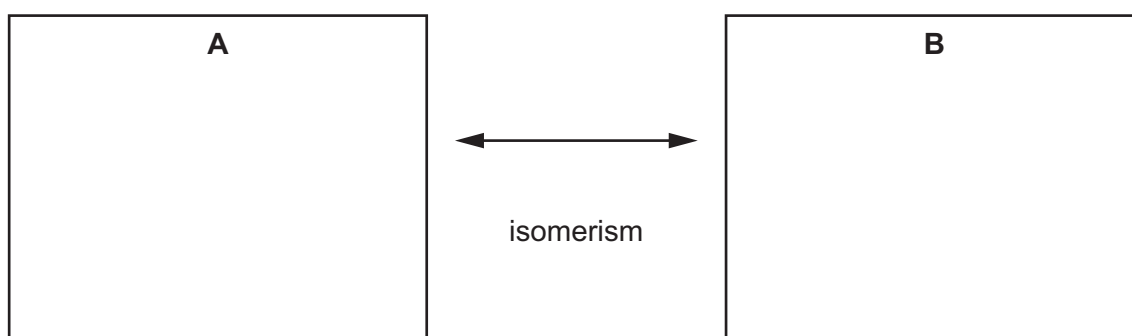
115. M/J 15/P21/Q4/a,b(ii,iii)

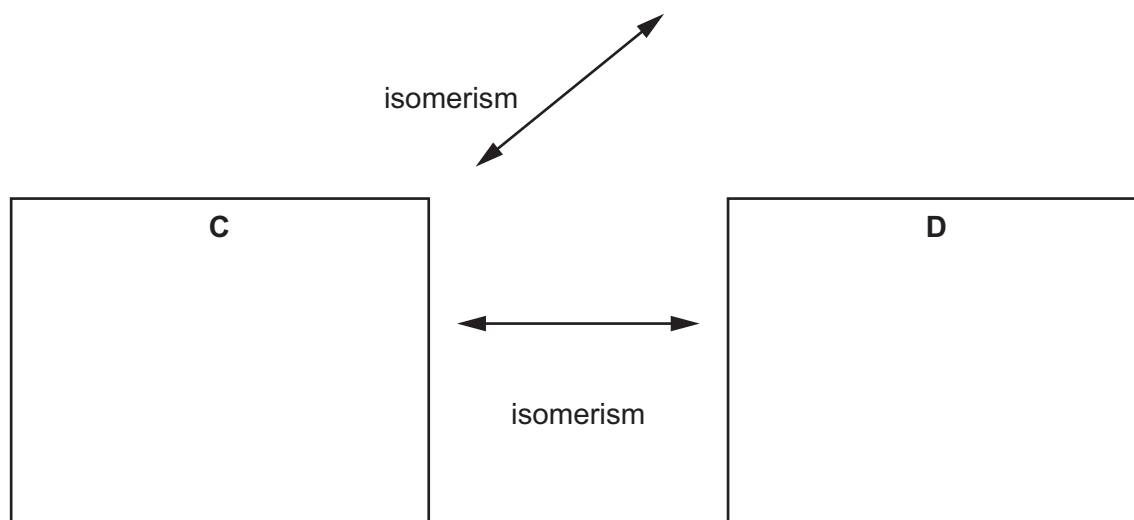
- 4** There are four alcohols, **A**, **B**, **C** and **D**, which are structural isomers with the molecular formula $C_4H_{10}O$.

Alcohol **A** does not react with acidified potassium dichromate(VI) solution but **B**, **C** and **D** do.

All four alcohols react with hot, concentrated sulfuric acid to form products with the molecular formula C_4H_8 . **A**, **C** and **D** each give a single product in this reaction. **B** gives a mixture of two structural isomers, one of which shows stereoisomerism.

(a) Give the **skeletal** formula for each of the four alcohols and complete the diagram with the names of the types of structural isomerism shown by each linked pair of compounds.



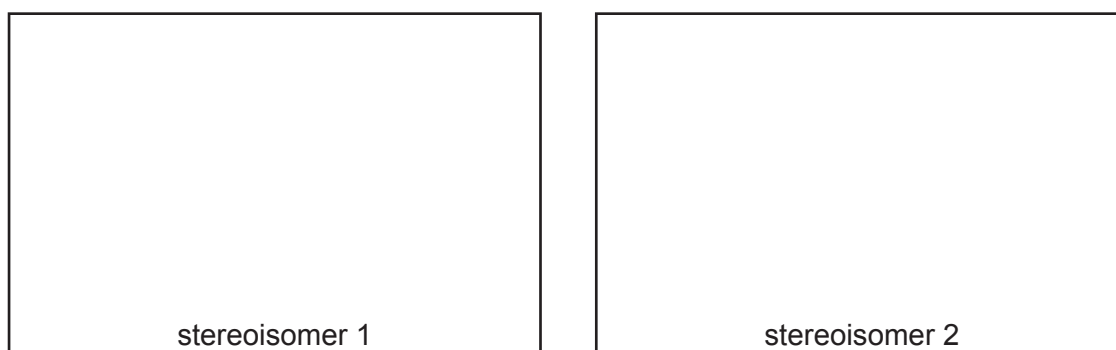


[7]

- (ii) State which of these two isomers shows stereoisomerism. Explain why this molecule is capable of showing stereoisomerism.

[2]

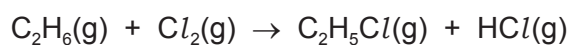
- (iii) Draw **displayed** formulae to show the two stereoisomers.



[2]

116. M/J 15/P22/Q3/b(i)

- 3** Ethane reacts with chlorine to form chloroethane.



- (b) Chloroethane can be converted back into ethane by a two-stage process via an intermediate compound, **X**.



- (i) Give the name of **X**.

[1]

117. M/J 15/P22/Q4/a

- 4** There are seven structural isomers with the molecular formula $C_5H_{10}O$ that are carbonyl compounds. Four of these are aldehydes.

These four aldehydes, **A**, **B**, **C** and **D**, have the following properties.

- Aldehyde **A** has a straight chain while **B**, **C** and **D** are branched.
- Aldehyde **B** is the only one of the four isomers with a chiral centre and it exists as a pair of optical isomers.
- Aldehyde **C** has two methyl groups in its structure but **D** has three.

(a) (i) Give the structure of each of the four isomers.

A	B
C	D

[4]

(ii) Draw the three-dimensional structures of the two optical isomers of **B**.

[2]

118. M/J 15/P23/Q3/a(ii),b(i)

3 **A, B, C, D, E** and **F** are all structural isomers with the molecular formula C_4H_8O .

(a) **A, B** and **C** all give an orange precipitate when treated with 2,4-DNPH but only **A** and **B** give a brick-red precipitate when warmed with Fehling's solution.

(ii) Name the type of structural isomerism shown by **A** and **B**.

[1]

(b) **D, E** and **F** all decolourise bromine and effervesce slowly with sodium metal.

E shows geometrical isomerism. Only **D** has a branched chain.

None of these isomers contains an oxygen atom bonded to a carbon atom involved in π bonding.

None of these isomers contains a chiral centre.

(i) Give the structures of **D, E** and **F**. Show the two stereoisomers of **E** and label the stereoisomerism shown.

D	
E	E
F	

[5]

Answers Section

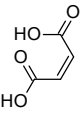
1. M/J 26/P21/Q5/a,c

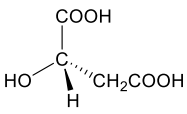
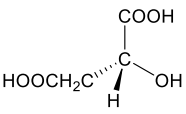
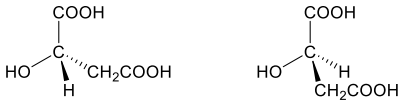
5(a)	<table><tr><th>compound</th><th>type of hybridisation of carbon atoms</th></tr><tr><td>1,2 dichloroethane</td><td>sp³</td></tr><tr><td>chloroethene</td><td>sp²</td></tr></table>	compound	type of hybridisation of carbon atoms	1,2 dichloroethane	sp ³	chloroethene	sp ²	2 M1 A sp ³ sp ³ SP ³ etc. M2 A sp ² sp ² SP ² etc.
compound	type of hybridisation of carbon atoms							
1,2 dichloroethane	sp ³							
chloroethene	sp ²							
5(c)	One atom takes both of the electrons when the covalent bond is broken ✓✓ M1 when a (covalent) bond / it is broken M2 one atom takes both of the electrons from the shared pair	2 A breaking = splitting = separation A breaking of a specific covalent bond e.g. when a CH bond is broken shared pair of electrons = covalent bond M2 A 'one atom takes both electrons' if reference to process involving breaking a (covalent) bond is described A takes = receives A electrons moves to / electrons transferred to I attract / pull towards M2 A species = atom R element, molecule, compound I radical I uneven number of electrons I formation of (positive and negative) ions I both electrons unequally distributed to one atom I formation of nucleophile and electrophile						

2. M/J 26/P22/Q5/a,b(i,ii)

5(a)	<table><tr><th>structure of product</th><th>systematic name of product</th></tr><tr><td>CH₃(CH₂)₅Cl</td><td>1-chlorohexane</td></tr><tr><td>CH₃(CH₂)₃CHClCH₃</td><td>2-chlorohexane</td></tr><tr><td>CH₃(CH₂)₃CH₂ClCH₂Cl</td><td>1,2 dichlorohexane</td></tr><tr><td>CH₃(CH₂)₁₀CH₃</td><td>dodecane</td></tr><tr><td>CH₃(CH₂)₃CH(CH₃)CH(CH₃)(CH₂)₃CH₃</td><td>5,6-dimethyldecane</td></tr></table>	structure of product	systematic name of product	CH ₃ (CH ₂) ₅ Cl	1-chlorohexane	CH ₃ (CH ₂) ₃ CHClCH ₃	2-chlorohexane	CH ₃ (CH ₂) ₃ CH ₂ ClCH ₂ Cl	1,2 dichlorohexane	CH ₃ (CH ₂) ₁₀ CH ₃	dodecane	CH ₃ (CH ₂) ₃ CH(CH ₃)CH(CH ₃)(CH ₂) ₃ CH ₃	5,6-dimethyldecane	2 Max 1 for consistently incorrect alkane stem that has correct idea of 1-chloro and 2-chloro. Max 1 6-chlorohexane and 5-chlorohexane. Max 1 for consistent errors e.g. 1-chlorohexane and 2-chlorohexane OR 1-chlorohexane and 2-chlorohexane OR chloro-1-hexane and chloro-2-hexane A missing hyphen
structure of product	systematic name of product													
CH ₃ (CH ₂) ₅ Cl	1-chlorohexane													
CH ₃ (CH ₂) ₃ CHClCH ₃	2-chlorohexane													
CH ₃ (CH ₂) ₃ CH ₂ ClCH ₂ Cl	1,2 dichlorohexane													
CH ₃ (CH ₂) ₁₀ CH ₃	dodecane													
CH ₃ (CH ₂) ₃ CH(CH ₃)CH(CH ₃)(CH ₂) ₃ CH ₃	5,6-dimethyldecane													
5(b)(i)	(species) with one or more unpaired electrons	1 BOD species with (an) unpaired electron(s) I lone / single / free / odd number of electrons (in valence shell) I species with specified number (except 1) unpaired electrons BOD species with one or two unpaired electrons R it is an unpaired electron BOD species = atom, ion, particle, molecule, substance, element, compound												
5(b)(ii)	homolytic (fission)	1 BOD homolytical A homolysis I covalent I photolytic												

3. M/J 26/P22/Q6/a,b(ii)

6(a)(i)		1 A any unambiguous cis / Z structure of P I connectivity I trans structure
---------	---	---

6(a)(ii)	M1 limited / restricted / lack of / no rotation about the pi bond / π bond of C=C M2 (two) different groups attached to each C of the C=C / each end of the C=C	2 M1 A limited / restricted / lack of / no rotation about the C=C reference to C=C needs to be made once only BOD C–C double bond I double bond (there is a C=O also in the molecule) I movement rotate = twist = spin R (compound) P has a chiral carbon M2 each = both A different atoms / functional groups / H and COOH / species I each side (of the C=C) R different compounds / molecules
6(b)(ii)	M1  M2 	2 COOH = CO₂H A C₂H₃O₂ M1 4 correct groups around central C in any orientation (might be seen on 'isomer 2') M2 enantiomer of their M1 structure (does not have to be shown as its mirror image).  M2 R bonds in the plane that are opposite to each other I bond shapes that are not associated with the chiral carbon. I bond angles I connectivity

4. M/J 26/P23/Q4/c(i,ii)

4(c)(i)	M1 (molecules / isomers with) the same molecular formula / same number of atoms of each element M2 different arrangement of atoms	2 A similar = same M1 A same molecular formula I same chemical formulas I same / similar molecular number M2 A different structures M2 A different structural formulas / different displayed formulas / different skeletal formulas I arrangement of bonds / different arrangements in space I different carbon skeleton / functional groups / chain
4(c)(ii)	M1 compound W AND compound V M2 both have (2) chiral centres / (2) asymmetric carbon atoms	2 M1 compound W AND compound V ✓ M2 one or two chiral centre on V & W ✓ M1 compound V OR compound W M2 show one or two chiral centre on the isomer identified ^✓ R molecules / compounds for atoms / groups R C attached to 4 different alkyl groups M2 A they have a chiral centre / an asymmetric carbon (atom) chiral = 4 different groups / atoms bonded to a carbon M2 may be indicated by annotation on Figure 4.2 e.g. CH₂OHCH₂*CH(OH) and / or *CH(OH)CH₂OH

5. M/J 26/P23/Q5/a(i)

5(a)(i)	structure of product	systematic name of product	2	A any unambiguous structures
	$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1 chloropropane		
	$\text{CH}_3\text{CHCl} / \text{CH}_3$	2-chloropropane		
	$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,4-dimethylbutane		
	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	hexane		
	$\text{CH}_3(\text{CH}_2)_2\text{CH}(\text{CH}_3)_2$	2-methylpentane		

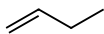
6. M/J 26/P24/Q5/c

5(c)	positional	1	A regoisomerism A position (isomer)
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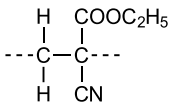
7. MAR 26/P22/Q2/a(i)

2(a)(i)	(free) radical	1
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8. MAR 26/P22/Q3/b

3(b)(i)		1
3(b)(ii)	M1  M2 <i>trans</i> -but-2-ene	2
3(b)(iii)	M1 (two) different groups attached to each carbon of C=C M2 restricted / lack of / no rotation about double / C=C / pi / π bond	2

9. MAR 26/P22/Q4/d

4(d)(i)	condensation	1
4(d)(ii)	methanal	1
4(d)(iii)	$\text{C}_6\text{H}_7\text{NO}_2$	1
4(d)(iv)		1
4(d)(v)	× on absorption at $\sim 1550\text{ cm}^{-1}$ - C=C (peak / C=C / bond) disappears (during / after polymerisation)	2

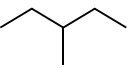
10. O/N 25/P21/Q3/b,c(i)

3(b)	two of: 	2
3(c)(i)	homolytic	1

11. O/N 25/P21/Q4/b(ii),c(i)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
4(c)(i)	2,3-dimethylbut-1-ene	1

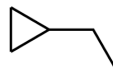
12. O/N 25/P22/Q3/a,d(i)

3(a)	M1 (family of molecules / compounds / series with) same general formula OR (family of molecules where) molecular formula differs (only) by (units of) $\text{CH}_{2(n)}$. M2 similar chemical properties / reactions	2
3(d)(i)	M1  M2 	2

13. O/N 25/P22/Q4/a(iii,iv,v)

4(a)(iii)	homolytic	1
4(a)(iv)	1,2-dichloropropane	1
4(a)(v)	M1 (type of structural isomerism) positional M2 (type of stereoisomerism) optical	2

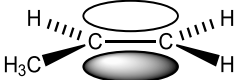
14. O/N 25/P23/Q3/b,c(i)

3(b)	two of:  OR  OR 	2
3(c)(i)	homolytic	1

15. O/N 25/P23/Q4/b(ii),c(i)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
4(c)(i)	2,3-dimethylbut-1-ene	1

16. O/N 25/P24/Q3/a(iv)

3(a)(iv)		1
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17. M/J 25/P22/Q3/a(i)

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	1
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18. M/J 25/P22/Q4/c(i)

4(c)(i)	sp^3	1
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19. M/J 25/P23/Q5/b(i)

5(b)(i)	number of σ bonds = 14 number of π bonds = 2	2
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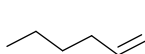
20. M/J 25/P23/Q6/a(i)

6(a)(i)	two or more reactants / molecules react to make (only) one product	1
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21. M/J 25/P24/Q4/a

4(a)	$C_5H_{10}O_2$	1
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22. M/J 25/P24/Q5/a,b,c

5(a)	M1  M2  OR 	2
5(b)(i)	no rotation / restricted rotation of C=C / pi bond	1
5(b)(ii)	(W / it has two) different groups bonded to each C of the double bond	1
5(c)		1

23. M/J 25/P24/Q6/a(iii)

6(a)(iii)	alkene(s)	1
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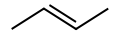
24. MAR 25/P22/Q1/c(i)

1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	1
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25. MAR 25/P22/Q4/b(iii)

4(b)(iii)	atom / carbon with four different atoms / (functional) groups attached / bonded	1
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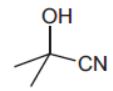
26. O/N 24/P21/Q1/d(ii,iii,iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

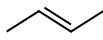
27. O/N 24/P21/Q2/b(i)

2(b)(i)	a species with one or more unpaired electrons	1
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28. O/N 24/P21/Q4/a(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms A 	1
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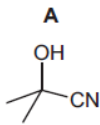
29. O/N 24/P23/Q1/d(ii,iii,iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

30. O/N 24/P23/Q2/b(i)

2(b)(i)	a species with one or more unpaired electrons	1
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31. O/N 24/P23/Q4/a(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms 	1
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32. M/J 24/P21/Q5/b,c(i)

5(b)(i)	sp^2	1
5(b)(ii)	M1 σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR  M2 π bond - diagram showing how π bond is made from sideways overlap of 2$\times$ p-orbitals i.e.  OR  OR 	2
5(c)(i)	EITHER (pair of) electrons in π bond are further away from the nuclei so weaker attraction OR (pair of) electrons in σ bond are closer to the two nuclei so stronger attraction	1

33. M/J 24/P21/Q6/a

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(a)(ii)	M1 number of stereoisomers = 8 M2 reasoning: 2 chiral centres AND 1 C=C (producing a cis-and trans or geometrical pair)	2
6(a)(iii)	$C_{13}H_{20}O_3$	1
6(a)(iv)	ester AND carbonyl / ketone AND C=C (bond) / alkene	1

34. M/J 24/P22/Q4/a,e

4(a)	Three-dimensional stereoisomers of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ M1 correct representation of left-hand answer as 3d structure of one of the enantiomers of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ M2 correct representation of 3d structure of the second enantiomer of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$	2			
4(e)(i)	 <table border="1" data-bbox="411 436 659 510"> <tr> <td>0</td> <td>2</td> <td>6</td> </tr> </table> 3 correct answers = 2 marks 2 correct answers = 1 mark 0 or 1 correct answers = 0 marks	0	2	6	2
0	2	6			
4(e)(ii)	M1 presence of a π bond M2 limited rotation of C=C bond M3 EITHER two different groups on each / both double bonded carbon atoms OR two different atoms on each / both double bonded carbon atom OR one H / atom and one alkyl group on each / both double bonded carbon atom	3			

35. M/J 24/P23/Q4/e(iii)

4(e)(iii)	butan-2-amine OR butyl-2-amine	1
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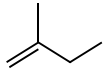
36. MAR 24/P22/Q4/a

4(a)	<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(x)</td><td>(x)</td><td>(x)</td></tr><tr><td>optical</td><td>(x)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	molecule	P	Q	R	S	geometrical	✓	(x)	(x)	(x)	optical	(x)	✓	✓	✓	1
	molecule	P	Q	R	S												
geometrical	✓	(x)	(x)	(x)													
optical	(x)	✓	✓	✓													
	<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(x)</td><td>(x)</td><td>(x)</td></tr><tr><td>optical</td><td>(x)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> CH₃CHBrCH₂CH₃	molecule	P	Q	R	S	geometrical	✓	(x)	(x)	(x)	optical	(x)	✓	✓	✓	1
molecule	P	Q	R	S													
geometrical	✓	(x)	(x)	(x)													
optical	(x)	✓	✓	✓													

37. O/N 23/P21/Q3/e(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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38. O/N 23/P21/Q4/a(i,ii,iii)

4(a)(i)	3-methylbut-1-ene	1
4(a)(ii)	one end / one C of the C=C bond has the same groups attached	1
4(a)(iii)		1

39. O/N 23/P22/Q2/e(i)

2(e)(i)	species with 1 or more unpaired electrons	1
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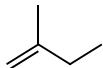
40. O/N 23/P22/Q4/a(i,ii)

4(a)(i)	(it has molecules that) contain a chiral carbon / centre OR are non-superimposable mirror images	1
4(a)(ii)	2-hydroxypropanoic acid	1

41. O/N 23/P23/Q3/e(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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42. O/N 23/P23/Q4/a(i,ii,iii)

4(a)(i)	3-methylbut-1-ene	1
4(a)(ii)	one end / one C of the C=C bond has the same groups attached	1
4(a)(iii)		1

43. M/J 23/P21/Q4/b(iii)

4(b)(iii)		type of hybridisation	sp^2	sp^3	2
		number of carbon atoms in V	3	1	

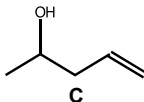
44. M/J 23/P22/Q3/a(ii),c

3(a)(ii)	propanenitrile			1
3(c)		type of bond in X	number of bonds in X	2
		sigma σ	8	
		pi π	2	

45. M/J 23/P22/Q4/d

4(d)				2
		sigma	pi	
	type of orbitals involved in bond	sp ² (-sp ²)	(2)p(-2)p)	
	how the orbitals overlap	direct	sideways	
	Two correct for one mark, four correct for two marks.			

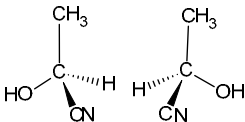
46. M/J 23/P22/Q5/a,c

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element AND different structural formulae / different structures	1
5(c)(i)	 M1 $CH_3CH(OH)$ present M2 five carbons in a straight chain AND molecular formula $C_5H_{10}O$ AND 1 C=C structure which shows only one type of stereoisomerism	2
5(c)(ii)	optical	1

47. M/J 23/P23/Q5/c

5(c)	type of hybridisation	sp^2	sp^3	2
	number of carbon atoms in X	2	4	

48. M/J 23/P23/Q6/c,d

6(c)(i)	(molecules / compounds with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(c)(ii)	M1 restricted rotation of C=C (bond) M2 each C of the double bond has two different groups / two different atoms (attached)	2
6(d)	 A pair of molecules of 2-hydroxypropanenitrile with bonds represented in 3-d AND the pair of molecules are non-superimposable images	1

49. MAR 23/P22/Q3/a(i,ii,iii),c

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) one product	1
3(a)(ii)	position(al isomerism)	1
3(a)(iii)	butanal	1
3(c)	alkane C_3H_8	1
	alcohols $C_4H_{10}O$	1

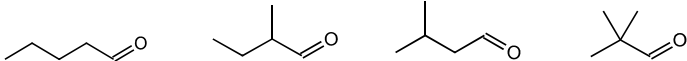
50. O/N 22/P22/Q3/a(ii,iii,iv)

3(a)(ii)	molecules with the same molecular formula AND different structural formulae <i>owtte</i>	1
3(a)(iii)	pentan-3-one	1
3(a)(iv)	<u>molecular</u> formula $C_5H_{10}O$	1

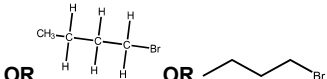
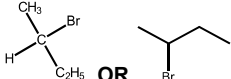
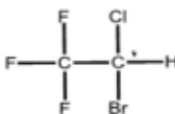
51. M/J 22/P21/Q3/a

3(a)	M1 optical M2 one of the C atoms has 4 different groups / atoms attached	2
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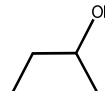
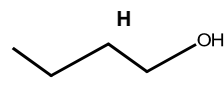
52. M/J 22/P21/Q4/c(iii)

4(c)(iii)	M1 / M2  M3 chain (isomerism)	3
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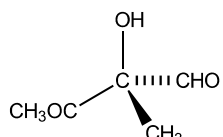
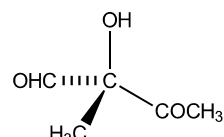
53. M/J 22/P22/Q4/a,b(i,ii)

4(a)(i)	$\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{Br}$  OR name: 1-bromobutane $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$  OR name: 2-bromobutane	2
	M1 structure of isomer 1 AND name M2 structure of isomer 2 AND name (in any order)	
4(a)(ii)	positional	1
4(b)(i)		1
4(b)(ii)	$\text{CF}_3\text{CBrClH} \rightarrow \text{CF}_3\text{CHCl}(\bullet) + \text{Br}(\bullet)$	1

54. M/J 22/P23/Q3/b

3(b)(i)	functional group	1
3(b)(ii)	positional	1
3(b)(iii)	 name: butan-2-ol  name: butan-1-ol	3
	M1 correct structures M2 correct structures in correct boxes H and J M3 correct corresponding name of both H and J	

55. MAR 22/P22/Q4/a(ii)

4(a)(ii)	 	2
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56. O/N 21/P21/Q3/c

3(c)(i)	$\text{C}_{18}\text{H}_{15}\text{P}$	1
3(c)(ii)	stage 1 = reduction stage 2 = substitution	2
3(c)(iii)	G = $\text{C}_2\text{H}_5\text{CH}_2\text{OH}$ H = $\text{C}_2\text{H}_5\text{CH}=\text{CHC}_2\text{H}_5$	2

57. O/N 21/P22/Q3/a,b,c,e(iv)

3(a)	(2,3-)dimethylbut-2-ene	1
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3(b)		1
3(c)(i)		1
3(c)(ii)	sp^2	1
3(e)(iv)	Draw two optical isomers of X. eg M1 One correct 3-dimensional representation of X M2 3-d structure which represents the enantiomer of their X	1
		1

58. O/N 21/P23/Q3/c

3(c)(i)	$C_{18}H_{15}P$	1
3(c)(ii)	stage 1 = reduction stage 2 = substitution	2
3(c)(iii)	G = $C_2H_5CH_2OH$ H = $C_2H_5CH=CHC_2H_5$	2

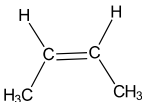
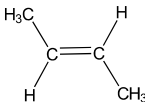
59. M/J 21/P21/Q5/a(i)

5(a)(i)	(compounds / molecules) containing only / entirely carbon and hydrogen (atoms)	1
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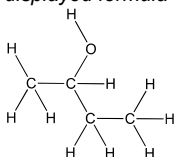
60. M/J 21/P21/Q6/c

6(c)		σ	π	2
	C_3H_6	8	1	
	C_3H_8O	11	0	

61. M/J 21/P22/Q4/a

4(a)(i)		structural formula	name	4
	A	$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$	but-1-ene OR 1-butene	
	B		cis / Z but-2-ene OR cis / Z 2-butene	
	C		trans / E but-2-ene OR trans / E 2-butene	
	D	$\text{CH}_2=\text{C}(\text{CH}_3)_2$	(2) methyl prop(-1-)ene	
	M1 correct identification of but-1-ene (for A) M2 correct identification of but-2-ene for B and C M3 identification of cis / trans (Z / E) for B and C (in correct order) M4 correct name of D			
4(a)(ii)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space			1

62. M/J 21/P22/Q5/a,b(i)

5(a)	displayed formula of butan-2-ol structure with O—H shown 	1
5(b)(i)	2-bromobutane	1

63. M/J 21/P23/Q4/a(i)

4(a)(i)	1,3-dichloropropan(-2-)one	1
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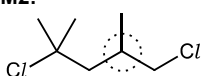
64. M/J 21/P23/Q5/c,d

5(c)	M1 chiral centre / non superimposable mirror image / has a C attached to 4 different groups	1
	M2 C=C where each C is attached to (two) different groups	1
5(d)	(molecules / isomers with) the same molecular formula / same number of atoms of each element AND different structural formulae / different structures	1

65. MAR 21/P22/Q4/a

4(a)	M1: $x = 108-110^\circ$ M2: $y = 118-122^\circ$	2
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66. O/N 20/P22/Q3/c(ii)

3(c)(ii)	M1: molecule with a non super-(im)posable mirror image	2
	M2: 	

67. O/N 20/P22/Q4/b(i)

4(b)(i)	positional (isomerism) / regioisomerism	1
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68. M/J 20/P21/Q5/b(ii)

5(b)(ii)	homolytic fission (of chlorine (gas) / Cl_2)	1
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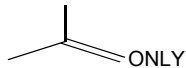
69. M/J 20/P21/Q6/a,g(i)

6(a)		1
6(g)(i)	Molecular formula of <i>W</i> $\text{C}_3\text{H}_6\text{O}_2$	1

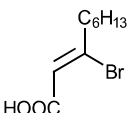
70. M/J 20/P22/Q3/a,b

3(a)	hydrolysis	1
3(b)	M1 both have molecular formula - $\text{C}_6\text{H}_{12}\text{O}_6$ M2 idea that in glucose and fructose there are the same number and type of atoms present but the atoms are arranged in a different order ie one has a carbonyl group at the end of the chain/molecule and the other has a carbonyl group in the middle of the chain/molecule	2

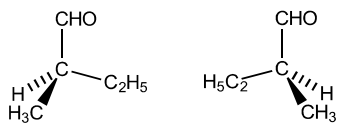
71. M/J 20/P22/Q4/c(iii,iv)

4(c)(iii)	 ONLY	1
4(c)(iv)	M1 2-chloro(-2-)methylpropane M2 1-chloro(-2-)methylpropane ALLOW in any order	2

72. MAR 20/P22/Q3/a(iv),c(i),d(i)

3(a)(iv)	it does not have four different (groups of) atoms attached to (central) carbon OR it does not have a chiral carbon / centre OR it has two identical / COOH groups attached to (central) carbon OR mirror image is super(im)posable	1
3(c)(i)	$\text{C}_5\text{H}_8\text{O}_2$	1
3(d)(i)	M1  M2 (two) different groups on each C atom in the $\text{C}=\text{C}$ / end of the $\text{C}=\text{C}$ double bond M3 no / restricted rotation about $\text{C}=\text{C}$	3

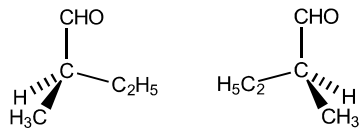
73. O/N 19/P21/Q4/a,c(iii,iv)

4(a)(i)	(2,2-)dimethylpropanal	1
4(a)(ii)	sp^2	1
4(c)(iii)	has a carbon / atom attached / bonded to four different atoms / groups / groups of atoms / chains	1
4(c)(iv)	 M1: Correct 3D representation M2: Correct 3D representation of drawn enantiomer	2

74. O/N 19/P22/Q4/d(i)

4(d)(i)	3-methylbutan-1-ol	1
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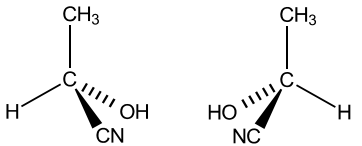
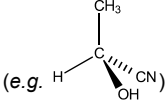
75. O/N 19/P23/Q4/a,c(iii,iv)

4(a)(i)	(2,2-)dimethylpropanal	1
4(a)(ii)	sp ²	1
4(c)(iii)	has a carbon / atom attached / bonded to four different atoms / groups / groups of atoms / chains	1
4(c)(iv)	 M1: Correct 3D representation M2: Correct 3D representation of drawn enantiomer	2

76. M/J 19/P21/Q4/a,b,c(i)

4(a)	3-chloroprop-1-ene	1
4(b)	$a = 109(.5)^\circ$	1
	$b = 120^\circ$	1
4(c)(i)	C ₃ H ₇ ClO ₂	1

77. M/J 19/P21/Q5/b

5(b)(i)	optical	1
5(b)(ii)	 M1 one 3-D structure of correct molecule shown.	1
	M2 a mirror image of the molecule drawn in M1 OR same profile with two groups swapped  (e.g. H-C(CH₃)(OH)(CN))	1
	M3 central chiral C shown as *	1

78. M/J 19/P22/Q1/a

1(a)(i)	C ₄ H ₁₀ / same molecular formula / OR same number of carbon (atoms) and hydrogen (atoms)	1
	different structural formula OR description of different structural formula which does not imply stereoisomerism	1
1(a)(ii)	structural / chain	1

79. M/J 19/P22/Q4/d(i,ii,v)

4(d)(i)	unpaired electron(s)	1
4(d)(ii)	homolytic (fission)	1
	one electron goes to each chlorine / atom	1
	OR pair of electrons is shared out (equally between the two atoms)	
4(d)(v)	M1 identify bond types in alkanes and alkenes bonds in alkanes are (all) sigma / σ AND alkenes have (C = C made of) sigma AND pi / σ AND π	1
	M2 electrons in π (of the C = C) are responsible for the reaction electrophiles are attracted to / attack to electrons in pi / π OR electrophiles react with pi / π	1

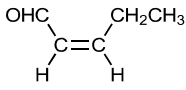
80. M/J 19/P22/Q5/c(i)

5(c)(i)	C ₃ H ₆ O ₂	1
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81. M/J 19/P23/Q6/a(iii)

6(a)(iii)	C ₄ H ₈	1
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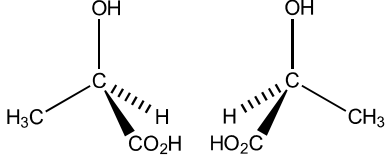
82. MAR 19/P22/Q3/c

3(c)	M1 geometric(al) M2 	2
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83. MAR 19/P22/Q4/a

4(a)	3-chloroprop-1-ene	1
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84. O/N 18/P22/Q4/b

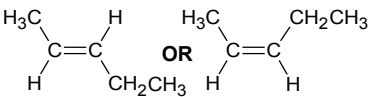
4(b)(i)	(Molecules that are) non-super(im)posable mirror images	1
4(b)(ii)	 M1 correct 3-d drawing of one isomer of Q [1] M2 correct pair of 3-d structures of the optical isomers of Q [1]	2

85. M/J 18/P22/Q4/a(ii),b,d(ii,iii)

4(a)(ii)	butanone	1
4(b)	CH ₃ CH ₂ CH ₂ CHO	1
	(CH ₃) ₂ CHCHO	1

4(d)(ii)	optical	1
4(d)(iii)	(X has a) chiral centre / asymmetric carbon atom OR (X has a) C atom attached to four different groups / atoms / chains	1
	non-super(im)posable mirror images	1

86. M/J 18/P23/Q2/a,b(i,iii)

2(a)(i)	(molecules / isomers with) the same molecular formula / same number of atoms of each element	1
	different structural formulae / different structures	1
2(a)(ii)	(Molecules / isomers) with the same (molecular and) structural formula	1
	different arrangement of atoms in space / different spatial arrangement of atoms.	1
2(b)(i)	two Hs on one of the C=C carbons / terminal C / C-1	1
	no chiral C / no C with 4 different groups / atoms / chains attached has a super(im)posable mirror image	1
2(b)(iii)	structure 	1
	trans-pent-2-ene or <i>E</i> - or cis-pent-2-ene or <i>Z</i> -	1

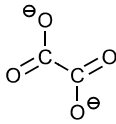
87. M/J 18/P23/Q4/a(i),b

4(a)(i)	2-bromobutane	1
4(b)	$(\text{CH}_3)_3\text{CBr}$ $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$ $\text{CH}_2\text{BrCH}_2\text{CH}_2\text{CH}_3$ 3 correct = 2 marks 2 correct = 1 mark	2

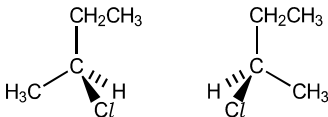
88. MAR 18/P22/Q2/b(i)

2(b)(i)	(a molecule / compound that is made up of) carbon and hydrogen (atoms) only	1
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89. MAR 18/P22/Q3/b(ii)

3(b)(ii)	 displayed structure of ethanedioate two – charges on carboxylates OR 2– charge overall	2
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90. O/N 17/P21/Q3/c(i),d(i,ii,iv,v)

3(c)(i)	(different molecules) with same molecular formula / same numbers of atoms of (each type) of element	1
	different structural formulae / displayed formulae	1
	chain / skeletal	2
	functional group	
	position(al) / regioisomerism	
	two types correct = 1 mark, all three correct = 2 marks	
3(d)(i)	(different molecules) with the same (molecular and) structural formula /	1
	with different arrangements of <u>atoms</u> in space / spatial arrangement of <u>atoms</u>	1
3(d)(ii)	mirror images are super(im)posable / no chiral carbon / no chiral centre / it is achiral	1
	(one) C of double bond has identical groups / H (atoms) (attached) OR (one) end of double bond has identical groups / 2 H (atoms) (attached)	1
3(d)(iv)	optical (isomerism)	1
3(d)(v)	one acceptable 3D structure of 2-chlorobutane	1
	the 2nd optical isomer EITHER drawn as a mirror image of the first OR the same bond pattern is shown but two of the groups swap positions.	1
		

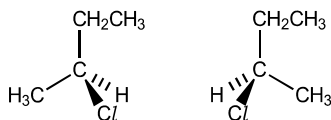
91. O/N 17/P22/Q4/c(i),d(i)

4(c)(i)	(different molecules) same molecular formula / same numbers of atoms of each (type of) element	1
	different structural formulae / displayed formulae	1
	chain / skeletal	2
	functional group	
	position(al) / regioisomerism	
	two types correct = 1 mark, all three correct = 2 marks	
4(d)(i)	mirror images are super(im)posable OR not chiral / no chirality / no chiral/asymmetric carbon/centre / achiral	1
	one or both C/end of double bond has identical groups / 2 methyl groups / 2 H (atoms)	1

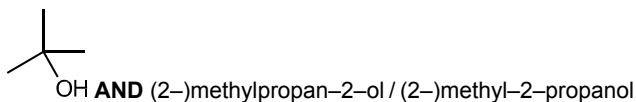
92. O/N 17/P23/Q3/c(i),d(i,ii,iv,v)

3(c)(i)	(different molecules) with same molecular formula / same numbers of atoms of (each type) of element	1
	different structural formulae / displayed formulae	1
	chain / skeletal	2
	functional group	
	position(al) / regioisomerism	
	two types correct = 1 mark, all three correct = 2 marks	
3(d)(i)	(different molecules) with the same (molecular and) structural formula /	1
	with different arrangements of <u>atoms</u> in space / spatial arrangement of <u>atoms</u>	1
3(d)(ii)	mirror images are super(im)posable / no chiral carbon / no chiral centre / it is achiral	1
	(one) C of double bond has identical groups / H (atoms) (attached) OR (one) end of double bond has identical groups / 2 H (atoms) (attached)	1

3(d)(iv)	optical (isomerism)	1
3(d)(v)	one acceptable 3D structure of 2-chlorobutane	1
	the 2nd optical isomer EITHER drawn as a mirror image of the first OR the same bond pattern is shown but two of the groups swap positions.	1

**93. M/J 17/P21/Q1/b(iv)**

1(b)(iv)	 OH AND propan-2-ol / 2-propanol	1
	 AND propan-1-ol / 1-propanol	1
	Alternative answers (any two): OH AND butan-1-ol / 1-butanol	
	 OH AND butan-2-ol / 2-butanol	
	 AND (2-)methylpropan-1-ol / (2-)methyl-1-propanol	

**94. M/J 17/P21/Q4/a,d(i)**

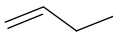
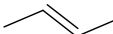
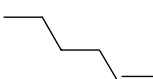
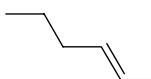
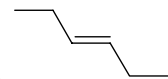
4(a)(i)	(molecules / isomers with) the same molecular formula / same number of atoms of each element	1
	different structural / displayed formulae / arrangement of bonds	1
4(a)(ii)	sp ² overlap of (2)s with two (2)p (atomic) orbitals	1
	sp ³ overlap of (2)s with all three (2)p (atomic) orbitals	1
4(a)(iii)	sp ² = 116° – 124°	1
	sp ³ = 106° – 112°	1
4(d)(i)	(molecules / isomers with) the same (molecular and) structural formula	1
	Any two of: chiral centre / C attached to four different groups / atoms non-super(im)posable mirror images different spatial / 3D arrangement of atoms (owtte) different rotation of plane-polarised light	1

95. M/J 17/P22/Q4/a(i,iii,iv),c(iii)

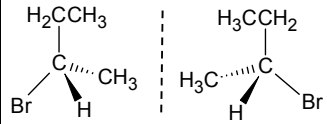
4(a)(i)	(A =)	1
4(a)(iii)	CH ₃ CHCHCH ₃ OR CH ₃ CH=CHCH ₃	1
4(a)(iv)	No rotation / restricted / limited rotation of C=C / (carbon) double bond	1
	One (of the two) methyl groups / one (of the two) H (atoms) is on each C (of C=C)	1

4(c)(iii)	a (tetrahedral) atom with four different groups / atoms / substituents attached OR a carbon (atom) with four different groups / atoms / substituents attached	1
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96. M/J 17/P23/Q1/a,b(iv)

1(a)	(molecules / isomers with) the same molecular formula / same number of atoms of each element	1
	different structural / displayed formulae / different arrangement of bonds	1
1(b)(iv)		1
		1
	alternative using supplied data: any two   	

97. M/J 17/P23/Q4/b,c(i),d(i,ii,iii),e(i)

4(b)	In any order $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ / $\text{CH}_2\text{CHCH}_2\text{CH}_3$ / $\text{CH}_2\text{CHC}_2\text{H}_5$ AND $\text{CH}_3\text{CH}=\text{CHCH}_3$ / $\text{CH}_3\text{CHCHCH}_3$ AND $(\text{CH}_3)_2\text{C}=\text{CH}_2$ / $(\text{CH}_3)_2\text{CCH}_2$	1
4(c)(i)	(different) molecules with the same (molecular and) structural formula	1
	(due to) different arrangement in space caused by $\text{C}=\text{C}$ / double bond	1
4(d)(i)	a (tetrahedral) atom with four different groups / atoms / substituents attached OR a carbon (atom) with four different groups / atoms / substituents attached	1
4(d)(ii)	but-1-ene	1
4(d)(iii)	 One 3D structure of 2-bromobutane which must have 2 bonds shown the same and two different, i.e. three bond types altogether, e.g. two solid lines, one wedge and one dash. If two bonds are drawn in the plane of the paper, i.e. single solid lines, they must not be at 180 degrees to each other. Second structure either mirror of first OR all bonds drawn the same with position of two groups swapped.	1
4(e)(i)	(2-)methylpropene / (2-)methylprop-1-ene	1

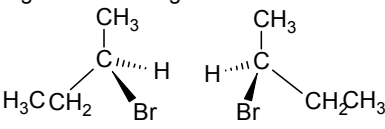
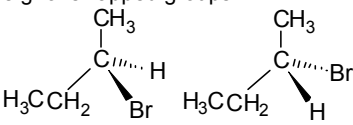
98. O/N 16/P21/Q4/a,b(i),c(i)

4(a)(i)	4-methylhex-2-ene	1	1
4(a)(ii)	(Molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms/groups (in space)	1	1
4(a)(iii)	4	1	4
	double-bond / alkene	1	
	(2) different groups on each double-bonded carbon	1	
	(one) chiral carbon (centre) / (one) carbon atom has 4 different groups attached / is asymmetric / is chiral	1	
4(b)(i)	2,3-dimethylbut-2-ene	1	1
4(c)(i)	(2-)methylprop(-1-)ene	1	1

99. O/N 16/P21/Q5/a

5(a)		1	1
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100. O/N 16/P22/Q4/a(ii,iii)

4(a)(ii)	e.g. of mirror images  e.g. of swapped groups 	1+1	
4(a)(iii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$ $(\text{CH}_3)_3\text{CBr}$	1 1 1	

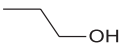
101. O/N 16/P22/Q5/a(ii)

5(a)(ii)	Cl^- and $^+\text{CH}_3/\text{CH}_3^+$	1	
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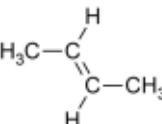
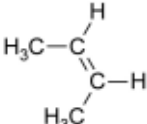
102. O/N 16/P23/Q4/a,b(i),c(i)

4(a)(i)	4-methylhex-2-ene	1	1
4(a)(ii)	(Molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms/groups (in space)	1	1
4(a)(iii)	4 double-bond / alkene (2) different groups on each double-bonded carbon (one) chiral carbon (centre) / (one) carbon atom has 4 different groups attached / is asymmetric / is chiral	1 1 1 1	4
4(b)(i)	2,3-dimethylbut-2-ene	1	1
4(c)(i)	(2-)methylprop(-1-)ene	1	1

103. O/N 16/P23/Q5/a

5(a)		1	1
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104. M/J 16/P21/Q4/a,b,c,d

4 (a)	$\text{CH}_2=\text{CHCH}_2\text{CH}_3 / \text{CH}_2\text{CHCH}_2\text{CH}_3$ AND $\text{CH}_3\text{CH}=\text{CHCH}_3 / \text{CH}_3\text{CHCHCH}_3$	[1]	[1]
(b)	$\text{CH}_2=\text{CHCH}_2\text{CH}_3 / \text{CH}_2\text{CHCH}_2\text{CH}_3$ AND $(\text{CH}_3)_2\text{C}=\text{CH}_2 / (\text{CH}_3)_2\text{CCH}_2$	[1]	[1]
(c)	  <i>trans</i>-but-2-ene (or <i>E</i>) <i>cis</i>-but-2-ene (or <i>Z</i>)	[1] [1]	[2]

(d)	B is $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ OR $\text{CH}_3\text{CH}=\text{CHCH}_3$ OR $(\text{CH}_3)_2\text{C}=\text{CH}_2$ distinguished by addition of bromine brown/red/orange/yellow to colourless/decolourises with B (but not A)	[1] [1] [1]	[3]
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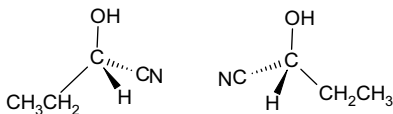
105. M/J 16/P21/Q5/e(ii)

(ii)	V = geometric(al) / <i>cis-trans</i> / <i>E-Z</i> T = optical	[1] [1]	[2]
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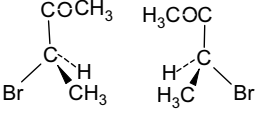
106. M/J 16/P22/Q4/a,b

4 (a)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ $(\text{CH}_3)_2\text{CHCOOH}$ / $\text{CH}_3\text{CH}(\text{CH}_3)\text{COOH}$	[1] [1]	[2]
(b) (i)	Two from 1. $\text{CH}_3\text{CH}_2\text{COOCH}_3$ 2. $\text{CH}_3\text{COOCH}_2\text{CH}_3$ 3. $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$	[1] [1]	[2]
(ii)	correct acid + alcohol for either ester 1. methanol + propanoic acid 2. ethanol + ethanoic acid 3. propan-1-ol + methanoic acid (conc) H_2SO_4 / (conc) H_3PO_4 AND heat/warm/reflux	[1] [1]	[2]

107. M/J 16/P22/Q5/c(ii)

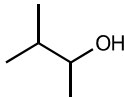
(ii)		[1+1]	[2]
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108. M/J 16/P23/Q4/a,d

4 (a)	3-hydroxybutan(-2-)one	[1]	[1]
(d) (i)	$\text{CH}_3\text{COCH}(\text{=})\text{CH}_2$	[1]	[1]
(ii)	one of the double-bonded C atoms / first C has 2H atoms attached ora so no <i>cis-trans</i> / <i>E-Z</i> / geometric(al) isomerism possible OR no chiral C so mirror images superimposable / molecule not asymmetric	[1] [1]	[2]
(iii)	asymmetric / chiral C atom / carbon with four different groups / atoms attached	[1]	[1]
(iv)		[1+1]	[2]

109. MAR 16/P22/Q4/a

4 (a) (i)	<u>C_4H_{10}</u>	[1]	[1]
(ii)	<u>C_4H_9</u>	[1]	[1]

(iii)		[1]	[1]
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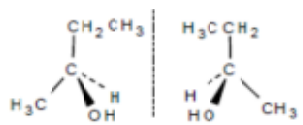
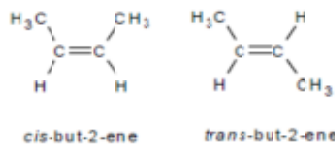
110. MAR 16/P22/Q5/a(ii)

(ii)	pent-3-en(e)-2-one OR 3-penten-2-one	[1]	[1]
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111. O/N 15/P21/Q3/a

3 (a) (i)	structural isomers: (different molecules with) same molecular formula but different structural formulae chiral: has a carbon /C attached to 4 different groups /atoms/ chains OR has no plane /line of symmetry/ has non-superimposable mirror images	[1] [1]	[2]
(ii)	CH ₃ CH ₂ CH(CH ₃)CH ₂ CH ₂ CH ₃ 3-methylhexane CH ₃ CH(CH ₃)CH(CH ₃)CH ₂ CH ₃ / (CH ₃) ₂ CHCH(CH ₃)CH ₂ CH ₃ 2,3-dimethylpentane	[1] [1] [1] [1]	[4]

112. O/N 15/P22/Q4/a(ii,iii,v,vi)

(ii)	Has a chiral centre /carbon OR has a <u>carbon /C</u> attached to 4 different groups /atoms/ chains OR has no plane /line of symmetry	[1]	[1]
(iii)		[1+1]	[2]
(v)	 <i>cis-but-2-ene</i> <i>trans-but-2-ene</i>	[1] [1]	[2]
(vi)	But-1-ene 2 Hs on one of the double-bonded Cs OR does not have 2 different groups on both atoms /each atom in C=C	[1] [1]	[2]

113. O/N 15/P23/Q3/a

3 (a) (i)	structural isomers: (different molecules with) same molecular formula but different structural formulae chiral: has a carbon /C attached to 4 different groups /atoms/ chains OR has no plane /line of symmetry/ has non-superimposable mirror images	[1] [1]	[2]
(ii)	CH ₃ CH ₂ CH(CH ₃)CH ₂ CH ₂ CH ₃ 3-methylhexane CH ₃ CH(CH ₃)CH(CH ₃)CH ₂ CH ₃ / (CH ₃) ₂ CHCH(CH ₃)CH ₂ CH ₃ 2,3-dimethylpentane	[1] [1] [1] [1]	[4]

114. M/J 15/P21/Q3/b(i)

(b) (i)	Stereoisomerism = (molecules with the same molecular formula and same structural formula but different spatial arrangements of atoms)	[1]	
	Chiral centre = atom with four different atoms/groups attached	[1]	[2]

115. M/J 15/P21/Q4/a,b(ii,iii)

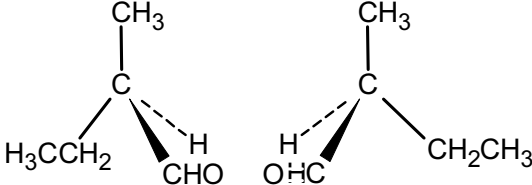
4 (a)	Diagram illustrating isomerism between four alcohols (A, B, C, D): - A and B are chain isomers. - C and D are chain isomers. - A and C are position isomers. - B and D are chain isomers. - C and B are chain OR position isomers. OR Diagram illustrating isomerism between four alcohols (A, B, C, D): - A and B are chain OR position isomers. - C and D are chain isomers. - C and B are chain OR position isomers.	[1] [1] [1] [1] [1] [1] [1]	[7]
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(ii)	but-2-ene AND two different groups on each carbon (of C=C) double bond means no free rotation	[1] [1]	[2]
(iii)	and (either way round)	[1+1]	[2]

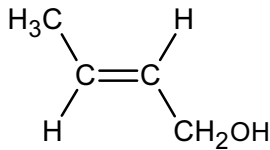
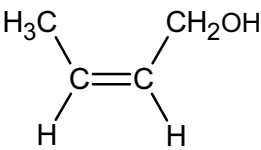
116. M/J 15/P22/Q3/b(i)

(b) (i)	ethene	[1]	[1]
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117. M/J 15/P22/Q4/a

4 (a) (i)	A = $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$ B = $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHO}$ C = $(\text{CH}_3)_2\text{CHCH}_2\text{CHO}$ D = $(\text{CH}_3)_3\text{CCHO}$	[1] [1] [1] [1]	[4]
(ii)		[1+1]	[2]

118. M/J 15/P23/Q3/a(ii),b(i)

(ii)	Chain	[1]	[1]
(b) (i)	D $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{OH}$	[1]	
	E  trans OR E	[1+1]	
	E  cis OR Z	[1]	
	F $\text{H}_2\text{C}=\text{CHCH}_2\text{CH}_2\text{OH}$	[1]	
			[5]